Bureau Veritas

Thruster Requirements

Comprehensive Surveyor Catalog

Based on NR467 Rules for Classification of Steel Ships

Sections 7, 14, and 15

Complete Plan Approval Reference Guide

Document Version 1.2
2026

FOR BUREAU VERITAS PLAN APPROVAL SURVEYORS

Contents

Document Purpose and Usage

This catalog provides a comprehensive overview of thruster requirements per Bureau Veritas Rules for Classification (NR467) and related guidance, designed as a practical reference for plan approval surveyors.

Key Features

- Complete drawing and document lists for all thruster types
- Explicit approval categories (**FA** or **FI**)
- Cross-references for overlapping requirements (steering gear, main shafting, hydraulics)
- Surveyor checklists by thruster type
- Testing and commissioning requirements
- Common submission errors and review tips

Document Structure

The catalog is organized into the following chapters:

1. General Requirements (All Thrusters >110kW)
2. Transverse Thrusters (Tunnel & Retractable)
3. Rotating and Azimuth Thrusters (incl. Podded Units)
4. Water-Jet Propulsion Systems
5. Voith-Schneider (Cycloidal) Propulsion Systems
6. Other Thruster Types (Rim-Driven, Pump-Jet, Gill/Jet, Steering Grid)
7. Prime Mover Considerations — Diesel-Engine vs. Electric-Motor Driven Thrusters
8. Thrusters as Part of Steering Gear
9. Thrusters as Part of Main Shafting
10. Hydraulic Systems & Pitch Control
11. Dynamic Positioning (DP) Thrusters
12. Ice Class & Cold-Climate (Winterisation) Requirements

How to Use This Catalog (Surveyor Workflow)

1. **Identify the thruster type** (Chapters 2–6) and confirm rated power. Thrusters >110 kW attract the full Sec. 15 submission set.
2. **Identify the prime mover** — electric motor or diesel engine (Chapter 7) — and add the corresponding drive-train drawings.
3. **Identify the functional role(s)**: propulsion, steering (Chapter 8), main shaft line (Chapter 9), or dynamic positioning (Chapter 11). Each role *adds* to the base package.
4. **Overlay additional notations & service constraints**: ice class / winterisation (Chapter 12), other BV additional class notations (Chapter 13), and the ship-type / service-notation requirements incl. SRtP, hazardous-area and offshore rules (Chapter 14).
5. **Verify hydraulics/pitch** (Chapter 10) and **testing & commissioning** (Chapter 15).
6. **Run the type-specific checklist** (Chapter 16) and screen against common errors (Chapter 17).

Legend

- **FA** = **For Approval** — Must be reviewed and approved before construction/installation
- FI = **For Information** — Submitted for record, not requiring formal approval
- **During Construction** = Certificates/reports generated during manufacturing

Note on rule references: Thruster requirements are found in **NR467, Part C, Chapter 1, Section 15** ("Thrusters"); steering gear in **Section 14**; shaft lines and propellers in **Sections 7–8**. Bracketed sub-references such as [2.3] are structural pointers within this catalog. Ice class, comfort, automation and DP requirements sit in **NR467 Part E** (Additional Class Notations) and dedicated Rule Notes (NR216, NR527, etc.). Always confirm the exact article against the rule edition applicable to the vessel's contract/build date.

Chapter 1

General Requirements - All Thrusters >110kW

1.1 Applicability

All thrusters with power output exceeding 110 kW must comply with NR467 Section 15 requirements. Thrusters $\leq$110 kW may have reduced submission requirements (verify with surveyor).

1.2 Mandatory Submissions for All Thruster Types

Table 1.1: General Requirements - All Thrusters

#	Drawing/Document	Category	Content Requirements	Reference
1	General Arrangement Drawing	**FA**	• Thruster location on vessel • Installation orientation • Access provisions • Clearances to other equipment • Coordinate system reference	NR467 Sec.15, [1]
2	Technical Specification Sheet	**FA**	• Rated power (kW) • Rated thrust (kN) • Rated speed (RPM) • Operating voltage/pressure • Weight and dimensions • Operating depth/pressure rating	NR467 Sec.15, [1]

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Table 1.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
3	Material Specifications	**FA**	• Material grades for all major components • Compliance with NR467 Part B, Ch.3 • Corrosion protection details • Material certificates (during construction)	NR467 Part B, Ch.3
4	Welding Procedures	**FA**	• Welding Procedure Specifications (WPS) • Welder qualification records • NDT requirements and extent • Weld inspection plan	NR467 Part B, Ch.4
5	Manufacturer's Quality Plan	**FA**	• Manufacturing sequence • Inspection and test points • Quality control procedures • Traceability system	NR467 Sec.15
6	Operation & Maintenance Manual	**FI**	• Operating instructions • Maintenance schedule • Troubleshooting guide • Safety precautions	NR467 Sec.15
7	Spare Parts List	**FI**	• Recommended spare parts • Critical component identification • Part numbers and suppliers	NR467 Sec.15
8	Installation Manual	**FI**	• Installation procedure • Alignment requirements • Foundation specifications • Connection details	NR467 Sec.15

Chapter 2

Transverse Thrusters (Tunnel & Retractable)

2.1 Tunnel Thrusters (Fixed Installation)

2.1.1 Structural Drawings & Calculations

Table 2.1: Tunnel Thrusters - Structural Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Tunnel Structural Arrangement	**FA**	• Tunnel diameter and length • Tunnel position relative to baseline/centerline • Plate thickness and material • Reinforcement arrangement • Connection to hull structure	NR467 Sec.15, [2.1]
2	Tunnel Structural Calculations	**FA**	• Pressure loads (internal/external) • Structural stress analysis • Buckling analysis • Fatigue assessment • FEA results (if applicable)	NR467 Sec.15, [2.1]
3	Tunnel End Grating/Grid Design	**FA**	• Grating dimensions and material • Attachment method • Structural strength verification • Debris protection	NR467 Sec.15, [2.2]
4	Hull Penetration Details	**FA**	• Opening dimensions • Reinforcement around opening • Welding details • Watertight integrity provisions	NR467 Part B, Ch.5

2.1.2 Thruster Unit - Mechanical Components

Table 2.2: Tunnel Thrusters - Mechanical Components

#	Drawing/Document	Category	Content Requirements	Reference
5	Thruster Unit General Assembly	**FA**	• Complete unit cross-section • Component identification • Material specifications • Assembly sequence	NR467 Sec.15, [2.3]
6	Propeller Design Drawing	**FA**	• Propeller diameter and pitch • Number of blades • Blade profile and geometry • Hub design • Material specification	NR467 Sec.15, [2.4]
7	Propeller Strength Calculations	**FA**	• Blade root stress analysis • Hub stress analysis • Centrifugal force calculations • Fatigue life assessment • Cavitation analysis	NR467 Sec.15, [2.4]
8	Propeller Shaft Drawing	**FA**	• Shaft diameter and length • Material specification • Surface finish requirements • Keyway details • Seal locations	NR467 Sec.15, [2.5]
9	Shaft Strength Calculations	**FA**	• Torsional strength analysis • Bending stress calculations • Combined stress analysis • Deflection calculations • Critical speed analysis • Fatigue assessment	NR467 Sec.15, [2.5]; Sec.7

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Table 2.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
10	Bearing Arrangement Drawing	**FA**	• Bearing type and manufacturer • Bearing housing design • Lubrication arrangement • Seal design • Cooling provisions (if applicable)	NR467 Sec.15, [2.6]
11	Bearing Load Calculations	**FA**	• Radial and axial loads • Bearing life calculation (L10) • Lubrication requirements • Temperature analysis	NR467 Sec.15, [2.6]
12	Gearbox Assembly Drawing	**FA**	• Gear arrangement (bevel, helical, etc.) • Gear ratios • Casing design • Lubrication system • Mounting arrangement	NR467 Sec.15, [2.7]
13	Gear Tooth Calculations	**FA**	• Gear tooth strength (ISO 6336) • Contact stress analysis • Bending stress analysis • Lubrication film thickness • Efficiency calculations	NR467 Sec.15, [2.7]
14	Gearbox Lubrication System	**FA**	• Oil type and capacity • Circulation system (if forced) • Cooling arrangement • Filtration system • Oil level and temperature monitoring	NR467 Sec.15, [2.7]

2.1.3 Drive System (Electric or Hydraulic)

For Electric Motor Drive

Table 2.3: Tunnel Thrusters - Electric Drive System

#	Drawing/Document	Category	Content Requirements	Reference
15	Electric Motor Specification	**FA**	• Motor type (induction, synchronous) • Rated power and speed • Voltage and frequency • Insulation class • Protection class (IP rating) • Cooling method • Mounting arrangement	NR467 Sec.15, [2.8]; IEC 60092-301
16	Motor Foundation Drawing	**FA**	• Foundation structure design • Vibration isolation details • Bolt arrangement • Alignment provisions • Structural attachment to hull	NR467 Sec.15, [2.8]
17	Motor-Gearbox Coupling	**FA**	• Coupling type and manufacturer • Alignment tolerances • Torque capacity • Installation procedure	NR467 Sec.15, [2.8]

For Hydraulic Motor Drive

Table 2.4: Tunnel Thrusters - Hydraulic Drive System

#	Drawing/Document	Category	Content Requirements	Reference
18	Hydraulic Motor Specification	**FA**	• Motor type (axial piston, radial piston, gear) • Displacement and speed range • Rated pressure • Efficiency curves • Mounting details	NR467 Sec.15, [2.9]
19	Hydraulic System Schematic	**FA**	• Complete hydraulic circuit • Component symbols and identification • Pressure and flow ratings • See Chapter 10 for detailed requirements	NR467 Sec.15, [2.9]

2.1.4 Control System

Table 2.5: Tunnel Thrusters - Control System

#	Drawing/Document	Category	Content Requirements	Reference
20	Control System Block Diagram	**FA**	• Control architecture • Input/output signals • Control logic overview • Interface with ship systems	NR467 Sec.15, [2.10]; Part C, Ch.4
21	Control Panel Layout	**FA**	• Panel arrangement • Component locations • Wiring diagram • Protection devices	NR467 Sec.15, [2.10]
22	Electrical Single-Line Diagram	**FA**	• Power supply arrangement • Circuit breakers and protection • Emergency stop circuit • Monitoring and alarm systems	NR467 Sec.15, [2.10]; IEC 60092-504
23	Control Software Description	**FA**	• Software functionality • Control algorithms • Safety functions • Software validation documentation	NR467 Part C, Ch.4

2.1.5 Additional Calculations & Documentation

Table 2.6: Tunnel Thrusters - Additional Calculations

#	Drawing/Document	Category	Content Requirements	Reference
24	Thrust Calculation	**FA**	• Theoretical thrust calculation • Efficiency factors • Operating curves (thrust vs. speed) • Bollard pull prediction	NR467 Sec.15, [2.11]

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Table 2.6 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
25	Vibration Analysis	**FA**	• Natural frequency calculations • Forced vibration analysis • Resonance avoidance verification • Vibration isolation design	NR467 Sec.15, [2.12]
26	Noise Level Prediction	**FA**	• Airborne noise prediction • Structure-borne noise analysis • Compliance with IMO noise code • Noise reduction measures	NR467 Sec.15, [2.12]
27	Thermal Analysis	**FA**	• Heat generation calculations • Cooling requirements • Temperature rise predictions • Ventilation requirements	NR467 Sec.15, [2.13]

2.2 Retractable Thrusters

2.2.1 Additional Requirements Beyond Fixed Tunnel Thrusters

All submissions from Section 2.1 apply, PLUS the following:

Table 2.7: Retractable Thrusters - Additional Requirements

#	Drawing/Document	Category	Content Requirements	Reference
28	Retraction Mechanism Assembly	**FA**	• Complete mechanism design • Lifting/lowering system • Guide rail arrangement • Travel limits and stops • Position indicators	NR467 Sec.15, [2.14]
29	Hydraulic Cylinder Specifications	**FA**	• Cylinder bore and stroke • Operating pressure • Rod diameter and material • Seal specifications • Mounting arrangement	NR467 Sec.15, [2.14]
30	Retraction System Calculations	**FA**	• Lifting force requirements • Hydraulic pressure calculations • Cylinder sizing verification • Retraction/extension time calculations • Dynamic load analysis	NR467 Sec.15, [2.14]
31	Hull Opening & Closure Design	**FA**	• Opening dimensions • Closure plate/door design • Sealing arrangement (extended position) • Sealing arrangement (retracted position) • Locking mechanism	NR467 Sec.15, [2.15]; Part B, Ch.5
32	Hull Opening Structural Analysis	**FA**	• Local hull strength calculations • Reinforcement design around opening • Stress concentration analysis • Fatigue assessment • FEA results (typically required)	NR467 Sec.15, [2.15]; Part B, Ch.5

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Table 2.7 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
33	Watertight Integrity Documentation	**FA**	• Seal design and material • Pressure testing requirements • Leakage criteria • Compliance with subdivision requirements	NR467 Part B, Ch.5
34	Locking System Design	**FA**	• Mechanical locks (extended position) • Retracted position locks • Locking force calculations • Fail-safe provisions • Interlock system	NR467 Sec.15, [2.16]
35	Position Indication System	**FA**	• Position sensors (type and location) • Indication on bridge/control panel • Alarm system (intermediate positions) • Redundancy provisions	NR467 Sec.15, [2.16]
36	Emergency Lowering System	**FA**	• Emergency power source • Manual lowering provisions • Emergency procedure • Time to deploy	NR467 Sec.15, [2.17]
37	Hydraulic System for Retraction	**FA**	• HPU specifications • Control valve arrangement • Piping layout • Accumulator sizing (if applicable) • Emergency lowering circuit • See Chapter 10 for detailed requirements	NR467 Sec.15, [2.17]
38	Retraction/Extension Procedures	**FI**	• Step-by-step operating procedure • Pre-operation checks • Safety precautions • Troubleshooting	NR467 Sec.15, [2.18]

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Table 2.7 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
39	Emergency Recovery Procedures	**FI**	• Emergency lowering procedure • Manual operation instructions • Recovery from failure modes • Contact information for support	NR467 Sec.15, [2.18]

Chapter 3

Rotating and Azimuth Thrusters

3.1 Azimuth Thrusters (Fixed Pitch & Controllable Pitch)

3.1.1 General Arrangement & Structure

Table 3.1: Azimuth Thrusters - General Arrangement

#	Drawing/Document	Category	Content Requirements	Reference
1	General Arrangement Drawing	**FA**	• Thruster location on vessel • Installation orientation • Azimuth rotation range (typically 360°) • Clearance envelope (rotating) • Access for maintenance	NR467 Sec.15, [3.1]
2	Azimuth Unit Complete Assembly	**FA**	• Longitudinal and transverse sections • All major components identified • Material specifications • Assembly sequence • Lifting points	NR467 Sec.15, [3.1]
3	Hull Penetration & Foundation	**FA**	• Hull opening dimensions • Structural reinforcement • Foundation design • Watertight integrity details • Load transfer to hull structure	NR467 Sec.15, [3.2]; Part B, Ch.5

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Table 3.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
4	Hull Structure Calculations	**FA**	• Local hull strength analysis • Foundation load distribution • Stress analysis • Fatigue assessment • FEA results (typically required)	NR467 Sec.15, [3.2]; Part B, Ch.5

3.1.2 Propeller & Shaft System

For Fixed Pitch Propellers

Table 3.2: Azimuth Thrusters - Fixed Pitch Propeller

#	Drawing/Document	Category	Content Requirements	Reference
5	Propeller Design Drawing	**FA**	• Propeller diameter and pitch • Number of blades • Blade geometry (sections) • Hub design • Material specification • Blade attachment method	NR467 Sec.15, [3.3]
6	Propeller Strength Calculations	**FA**	• Blade stress analysis • Hub stress analysis • Centrifugal forces • Hydrodynamic forces • Fatigue life assessment • Cavitation analysis	NR467 Sec.15, [3.3]

For Controllable Pitch Propellers (ADDITIONAL to Fixed Pitch)

Table 3.3: Azimuth Thrusters - Controllable Pitch Mechanism

#	Drawing/Document	Category	Content Requirements	Reference
7	CP Hub Mechanism Assembly	**FA**	• Hub internal mechanism • Pitch control rod/linkage • Blade retention system • Bearing arrangement • Seal design	NR467 Sec.15, [3.4]

Continued on next page

Table 3.3 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
8	Blade Retention Calculations	**FA**	• Blade root connection strength • Centrifugal force analysis • Hydrodynamic moment analysis • Bolt/pin strength verification • Fatigue assessment	NR467 Sec.15, [3.4]
9	Pitch Control Mechanism	**FA**	• Hydraulic cylinder (in hub or external) • Control rod design • Linkage system • Pitch range and indicators • Fail-safe position	NR467 Sec.15, [3.4]; Ch.10
10	Pitch Control Hydraulic System	**FA**	• Hydraulic supply arrangement • Rotary joint/slip ring design • Control valve system • Pitch feedback system • See Chapter 10 for detailed requirements	NR467 Sec.15, [3.4]; Ch.10

3.1.3 Vertical Shaft & Column

Table 3.4: Azimuth Thrusters - Vertical Shaft & Column

#	Drawing/Document	Category	Content Requirements	Reference
11	Vertical Shaft Assembly	**FA**	• Shaft dimensions and material • Upper and lower bearing locations • Thrust bearing arrangement • Seal system • Coupling details	NR467 Sec.15, [3.5]; Sec.7
12	Vertical Shaft Calculations	**FA**	• Torsional strength • Bending stress (from propeller forces) • Combined stress analysis • Deflection calculations • Critical speed analysis • Fatigue assessment	NR467 Sec.15, [3.5]; Sec.7

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Table 3.4 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
13	Thrust Bearing Design	**FA**	• Bearing type and manufacturer • Load capacity • Lubrication system • Cooling arrangement • Monitoring provisions	NR467 Sec.15, [3.6]
14	Thrust Bearing Calculations	**FA**	• Axial load analysis • Bearing life calculation • Temperature rise analysis • Lubrication film thickness	NR467 Sec.15, [3.6]
15	Radial Bearing Arrangement	**FA**	• Upper and lower bearing design • Bearing housing • Lubrication system • Seal arrangement	NR467 Sec.15, [3.7]
16	Column Structure Design	**FA**	• Column/strut dimensions • Material specification • Hydrodynamic fairing • Structural strength • Corrosion protection	NR467 Sec.15, [3.8]
17	Seal System Design	**FA**	• Seal type and material • Seal arrangement (upper/lower) • Pressure rating • Leakage monitoring • Replacement procedure	NR467 Sec.15, [3.9]

3.1.4 Gear System

Table 3.5: Azimuth Thrusters - Gear System

#	Drawing/Document	Category	Content Requirements	Reference
18	Gear Train Assembly	**FA**	• Gear arrangement (bevel, helical, planetary) • Gear ratios • Gear casing design • Lubrication system • Access for inspection	NR467 Sec.15, [3.10]
19	Gear Tooth Strength Calculations	**FA**	• Tooth bending stress (ISO 6336) • Contact stress • Pitting resistance • Scuffing analysis • Efficiency calculations	NR467 Sec.15, [3.10]
20	Gear Lubrication System	**FA**	• Oil type and capacity • Circulation system (forced/splash) • Cooling arrangement • Filtration system • Oil level and temperature monitoring	NR467 Sec.15, [3.10]

3.1.5 Slewing (Azimuth) System

Table 3.6: Azimuth Thrusters - Slewing System

#	Drawing/Document	Category	Content Requirements	Reference
21	Slewing Bearing Design	**FA**	• Bearing type and manufacturer • Internal/external gear (if applicable) • Load capacity (axial, radial, moment) • Mounting arrangement • Lubrication provisions	NR467 Sec.15, [3.11]

Continued on next page

Table 3.6 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
22	Slewing Bearing Calculations	**FA**	• Load analysis (static and dynamic) • Bearing life calculation • Bolt preload calculations • Overturning moment analysis	NR467 Sec.15, [3.11]
23	Azimuth Drive Mechanism	**FA**	• Drive motor (electric or hydraulic) • Gear/pinion arrangement • Drive ratio • Brake system • Locking mechanism (if required)	NR467 Sec.15, [3.12]
24	Azimuth Drive Calculations	**FA**	• Torque requirements • Acceleration/deceleration analysis • Motor sizing verification • Brake sizing • Response time calculations	NR467 Sec.15, [3.12]
25	Position Feedback System	**FA**	• Position sensor type and location • Accuracy and resolution • Signal processing • Redundancy (if required for DP/steering)	NR467 Sec.15, [3.13]

3.1.6 Power Transmission

For Electric Drive

Table 3.7: Azimuth Thrusters - Electric Power Transmission

#	Drawing/Document	Category	Content Requirements	Reference
26	Electric Motor Specification	**FA**	• Motor type and rating • Voltage and frequency • Cooling method • Insulation and protection class • Mounting arrangement	NR467 Sec.15, [3.14]; IEC 60092-301

Continued on next page

Table 3.7 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
27	Power Cable Routing	**FA**	• Cable route from switchboard • Cable support and protection • Penetrations through bulkheads • Cable sizing calculations	NR467 Sec.15, [3.14]; IEC 60092-352
28	Slip Ring or Rotary Joint	**FA**	• Type and manufacturer • Current rating • Maintenance requirements • Cooling provisions	NR467 Sec.15, [3.14]

For Hydraulic Drive

Table 3.8: Azimuth Thrusters - Hydraulic Power Transmission

#	Drawing/Document	Category	Content Requirements	Reference
29	Hydraulic Motor Specification	**FA**	• Motor type and displacement • Rated pressure and speed • Efficiency curves • Mounting details	NR467 Sec.15, [3.15]
30	Hydraulic System Schematic	**FA**	• Complete hydraulic circuit • HPU specifications • Piping arrangement • Control valve system • See Chapter 10 for detailed requirements	NR467 Sec.15, [3.15]; Ch.10

3.1.7 Control System

Table 3.9: Azimuth Thrusters - Control System

#	Drawing/Document	Category	Content Requirements	Reference
31	Control System Architecture	**FA**	• Overall control system topology • Propulsion control • Azimuth control • Pitch control (CP thrusters) • Interface with ship systems	NR467 Sec.15, [3.16]; Part C, Ch.4
32	Bridge Control Console	**FA**	• Control panel layout • Joystick/lever arrangement • Display and indication • Alarm system • Emergency stop provisions	NR467 Sec.15, [3.16]
33	Local Control Station	**FA**	• Local control panel design • Control mode selection • Indication and monitoring • Interlock with bridge control	NR467 Sec.15, [3.16]
34	Control Software Description	**FA**	• Software architecture • Control algorithms • Safety functions • Software validation documentation • Version control	NR467 Part C, Ch.4
35	Electrical Single-Line Diagram	**FA**	• Power supply arrangement • Circuit breakers and protection • Emergency power provisions • Monitoring and alarm circuits	NR467 Sec.15, [3.16]; IEC 60092-504

3.1.8 Additional Calculations & Documentation

Table 3.10: Azimuth Thrusters - Additional Calculations

#	Drawing/Document	Category	Content Requirements	Reference
36	Thrust and Torque Calculations	**FA**	• Propeller thrust curves • Torque requirements • Power absorption • Efficiency analysis • Bollard pull prediction	NR467 Sec.15, [3.17]
37	Vibration Analysis	**FA**	• Natural frequency calculations • Forced vibration analysis • Resonance avoidance • Foundation vibration isolation • Acceptable vibration levels	NR467 Sec.15, [3.18]
38	Noise Analysis	**FA**	• Airborne noise prediction • Structure-borne noise analysis • Compliance with noise regulations • Noise reduction measures	NR467 Sec.15, [3.18]
39	Hydrodynamic Analysis	**FA**	• Propeller hydrodynamic design • Cavitation analysis • Efficiency optimization • Interaction effects with hull	NR467 Sec.15, [3.19]
40	Thermal Analysis	**FA**	• Heat generation in gears, bearings, seals • Cooling system capacity • Temperature rise predictions • Ventilation requirements	NR467 Sec.15, [3.20]

3.2 Podded Propulsion Units

Podded propulsion units include all requirements from Section 3.1 (Azimuth Thrusters), with the following ADDITIONAL or MODIFIED requirements:

3.2.1 Pod-Specific Structural Components

Table 3.11: Podded Propulsion - Pod Structure

#	Drawing/Document	Category	Content Requirements	Reference
41	Pod Housing Design	**FA**	• Pod shell structure • Material specification • Hydrodynamic fairing • Structural strength analysis • Corrosion protection system	NR467 Sec.15, [3.21]
42	Pod Housing Calculations	**FA**	• Pressure loading analysis • Structural stress analysis • Fatigue assessment • Impact resistance • FEA results	NR467 Sec.15, [3.21]
43	Strut Structure Design	**FA**	• Strut dimensions and material • Hydrodynamic profile • Connection to pod and hull • Internal cable/pipe routing • Access provisions	NR467 Sec.15, [3.22]
44	Strut Structural Calculations	**FA**	• Bending stress analysis • Torsional stress • Combined loading • Fatigue analysis • Vibration analysis	NR467 Sec.15, [3.22]

3.2.2 Electric Motor in Pod

Table 3.12: Podded Propulsion - In-Pod Motor

#	Drawing/Document	Category	Content Requirements	Reference
45	In-Pod Motor Design	**FA**	• Motor type (permanent magnet, induction) • Rated power and speed • Voltage and frequency • Insulation system (marine environment) • Rotor and stator assembly • Bearing arrangement	NR467 Sec.15, [3.23]; IEC 60092-301
46	Motor Thermal Design	**FA**	• Heat generation calculations • Cooling system design • Temperature limits • Thermal monitoring • Insulation life assessment	NR467 Sec.15, [3.23]
47	Motor Cooling System	**FA**	• Cooling method (water-cooled, oil-cooled) • Heat exchanger design • Coolant circulation system • Pump specifications • Temperature monitoring	NR467 Sec.15, [3.24]
48	Motor Mounting & Support	**FA**	• Motor mounting arrangement in pod • Alignment provisions • Vibration isolation • Thermal expansion accommodation	NR467 Sec.15, [3.23]

3.2.3 Power Supply Through Strut

Table 3.13: Podded Propulsion - Power Supply

#	Drawing/Document	Category	Content Requirements	Reference
49	Power Cable Routing	**FA**	• Cable route through strut • Cable support and protection • Cable entry into pod • Sealing arrangements • Cable sizing calculations	NR467 Sec.15, [3.25]; IEC 60092-352

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Table 3.13 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
50	Slip Ring or Rotary Transformer	**FA**	• Type and manufacturer • Current and voltage rating • Cooling provisions • Maintenance access • Reliability data	NR467 Sec.15, [3.25]
51	Power Distribution in Pod	**FA**	• Internal switchgear/junction boxes • Protection devices • Monitoring systems • Watertight/waterproof rating	NR467 Sec.15, [3.25]

3.2.4 Propeller Direct Drive

Table 3.14: Podded Propulsion - Direct Drive

#	Drawing/Document	Category	Content Requirements	Reference
52	Direct Drive Arrangement	**FA**	• Motor shaft to propeller connection • Coupling or direct mounting • Bearing arrangement • Seal system • Alignment procedure	NR467 Sec.15, [3.26]
53	Shaft Seal System	**FA**	• Seal type and material • Seal arrangement • Pressure rating • Leakage monitoring • Emergency sealing provisions	NR467 Sec.15, [3.26]

3.2.5 Pod-Specific Testing

Table 3.15: Podded Propulsion - Testing Requirements

#	Drawing/Document	Category	Content Requirements	Reference
54	Pressure Testing Procedure	**FA**	• Test pressure and duration • Test setup • Acceptance criteria • Leakage limits	NR467 Sec.15, [3.27]
55	Motor Insulation Testing	**FA**	• Insulation resistance tests • High voltage tests • Partial discharge tests • Test conditions and acceptance criteria	IEC 60092-301

Chapter 4

Water-Jet Propulsion Systems

4.1 General Arrangement

Table 4.1: Water-Jets - General Arrangement

#	Drawing/Document	Category	Content Requirements	Reference
1	Water-Jet General Arrangement	**FA**	• Installation location on vessel • Inlet and outlet positions • Nozzle orientation • Access for maintenance • Clearances	NR467 Sec.15, [4.1]
2	Hull Integration Drawing	**FA**	• Inlet duct design • Hull penetration details • Structural integration • Hydrodynamic considerations • Reinforcement arrangements	NR467 Sec.15, [4.2]; Part B, Ch.5

4.2 Water-Jet Unit Components

Table 4.2: Water-Jets - Unit Components

#	Drawing/Document	Category	Content Requirements	Reference
3	Water-Jet Unit Assembly	**FA**	• Complete unit cross-section • Pump assembly • Inlet duct • Nozzle and steering/reversing mechanism • Material specifications	NR467 Sec.15, [4.3]

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Table 4.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
4	Impeller Design	**FA**	• Impeller geometry • Number of blades • Material specification • Blade profile • Hub design	NR467 Sec.15, [4.4]
5	Impeller Strength Calculations	**FA**	• Blade stress analysis • Hub stress analysis • Centrifugal forces • Hydrodynamic forces • Fatigue assessment • Cavitation analysis	NR467 Sec.15, [4.4]
6	Pump Shaft Design	**FA**	• Shaft dimensions and material • Bearing locations • Seal arrangements • Coupling details	NR467 Sec.15, [4.5]
7	Pump Shaft Calculations	**FA**	• Torsional strength • Bending stress • Combined stress analysis • Deflection • Critical speed • Fatigue assessment	NR467 Sec.15, [4.5]; Sec.7
8	Bearing Arrangement	**FA**	• Bearing types and locations • Bearing housing • Lubrication system • Seal design • Cooling provisions	NR467 Sec.15, [4.6]
9	Bearing Load Calculations	**FA**	• Radial and axial loads • Bearing life calculation • Lubrication requirements • Temperature analysis	NR467 Sec.15, [4.6]

4.3 Inlet Duct System

Table 4.3: Water-Jets - Inlet Duct

#	Drawing/Document	Category	Content Requirements	Reference
10	Inlet Duct Design	**FA**	• Duct geometry and dimensions • Material specification • Structural design • Hydrodynamic profile • Debris protection (grating/screen)	NR467 Sec.15, [4.7]
11	Inlet Duct Calculations	**FA**	• Structural strength analysis • Pressure loading • Flow analysis • Cavitation assessment	NR467 Sec.15, [4.7]
12	Hull Penetration & Reinforcement	**FA**	• Opening dimensions • Reinforcement design • Watertight integrity • Structural attachment	NR467 Sec.15, [4.7]; Part B, Ch.5

4.4 Steering and Reversing Mechanism

Table 4.4: Water-Jets - Steering & Reversing

#	Drawing/Document	Category	Content Requirements	Reference
13	Steering Nozzle Design	**FA**	• Nozzle geometry • Deflection range • Actuator arrangement • Structural design • Material specification	NR467 Sec.15, [4.8]
14	Reversing Bucket Design	**FA**	• Bucket geometry • Deployment mechanism • Actuator system • Structural strength • Material specification	NR467 Sec.15, [4.9]

Continued on next page

Table 4.4 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
15	Steering/Reversing Actuators	**FA**	• Hydraulic cylinder specifications • Operating pressure • Force requirements • Position feedback • Emergency provisions	NR467 Sec.15, [4.10]
16	Steering/Reversing Calculations	**FA**	• Actuator force requirements • Structural strength of nozzle/bucket • Response time analysis • Hydraulic system sizing	NR467 Sec.15, [4.10]

4.5 Drive System

4.5.1 For Direct Drive from Main Engine

Table 4.5: Water-Jets - Direct Drive

#	Drawing/Document	Category	Content Requirements	Reference
17	Drive Shaft Arrangement	**FA**	• Shaft line from engine to water-jet • Bearing locations • Coupling arrangements • Alignment provisions	NR467 Sec.15, [4.11]; Sec.7
18	Drive Shaft Calculations	**FA**	• Torsional strength • Bending stress • Critical speed analysis • Alignment calculations • See Chapter 9 for full requirements	NR467 Sec.15, [4.11]; Sec.7

4.5.2 For Gearbox Drive

Table 4.6: Water-Jets - Gearbox Drive

#	Drawing/Document	Category	Content Requirements	Reference
19	Gearbox Assembly	**FA**	• Gear arrangement • Gear ratios • Casing design • Lubrication system • Mounting arrangement	NR467 Sec.15, [4.12]
20	Gear Calculations	**FA**	• Gear tooth strength • Contact stress • Bending stress • Lubrication analysis • Efficiency	NR467 Sec.15, [4.12]

4.6 Hydraulic System (for steering/reversing)

Table 4.7: Water-Jets - Hydraulic System

#	Drawing/Document	Category	Content Requirements	Reference
21	Hydraulic System Schematic	**FA**	• Complete hydraulic circuit • HPU specifications • Control valves • Piping arrangement • See Chapter 10 for detailed requirements	NR467 Sec.15, [4.13]; Ch.10

4.7 Control System

Table 4.8: Water-Jets - Control System

#	Drawing/Document	Category	Content Requirements	Reference
22	Control System Architecture	**FA**	• Control system topology • Steering control • Reversing control • Engine/pump speed control • Interface with ship systems	NR467 Sec.15, [4.14]; Part C, Ch.4

Continued on next page

Table 4.8 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
23	Bridge Control Console	**FA**	• Control panel layout • Joystick/lever arrangement • Display and indication • Alarm system • Emergency stop	NR467 Sec.15, [4.14]
24	Electrical Single-Line Diagram	**FA**	• Power supply • Protection devices • Monitoring circuits • Emergency power provisions	NR467 Sec.15, [4.14]; IEC 60092-504

4.8 Performance Calculations

Table 4.9: Water-Jets - Performance Calculations

#	Drawing/Document	Category	Content Requirements	Reference
25	Thrust and Performance Calculations	**FA**	• Thrust curves • Power absorption • Efficiency analysis • Speed predictions • Bollard pull	NR467 Sec.15, [4.15]
26	Cavitation Analysis	**FA**	• Cavitation inception speed • Cavitation patterns • Erosion risk assessment • Mitigation measures	NR467 Sec.15, [4.15]
27	Vibration Analysis	**FA**	• Natural frequency calculations • Forced vibration analysis • Resonance avoidance • Foundation design	NR467 Sec.15, [4.16]
28	Noise Analysis	**FA**	• Airborne noise prediction • Structure-borne noise • Compliance with regulations • Noise reduction measures	NR467 Sec.15, [4.16]

Chapter 5

Voith-Schneider (Cycloidal) Propulsion Systems

5.1 Description & Classification Basis

The Voith-Schneider Propeller (VSP), or cycloidal propeller, is a vertical-axis propulsor in which a set of vertical blades projects below a rotating casing (rotor). Each blade oscillates about its own vertical axis through a kinematic control gear linked to a movable control (steering) centre. By displacing the control centre, the unit produces thrust of *any magnitude in any horizontal direction* at constant rotor speed. A VSP therefore combines **propulsion and steering in a single unit**; the steering function makes **NR467 Part C, Ch 1, Sec 14 (Steering Gear) and SOLAS II-1 Reg. 29 requirements applicable** (see Chapter 8). Typical applications: Voith Water Tractors, double-ended ferries, buoy tenders, mine-countermeasure vessels.

VSP units are assessed under **NR467 Pt C, Ch 1, Sec 15** for the mechanical thruster elements, **Sec 14** for the steering/control function, and **Sec 7** where the drive forms part of the propulsion shaft line.

5.2 VSP-Specific Structural & Mechanical Components

Table 5.1: Voith-Schneider Propeller — Mechanical & Structural Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	VSP Unit General Assembly	**FA**	• Complete unit cross-section (rotor casing, blades, gear) • Number of blades and blade circle diameter • Rotor speed and orbit geometry • Material list and lifting arrangement	NR467 Sec.15, [5.1]

Continued on next page

Table 5.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
2	Blade (Wing) Design Drawing	**FA**	• Blade profile, chord, span • Blade shaft dimensions and material • Blade root / shaft connection • Surface finish and corrosion protection	NR467 Sec.15, [5.2]
3	Blade Shaft & Blade Strength Calculations	**FA**	• Hydrodynamic blade loads (thrust + side force) • Bending + torsion of blade shaft • Centrifugal loads at max rotor speed • Fatigue assessment (reversing loads)	NR467 Sec.15, [5.2]; Sec.7
4	Blade Bearing Arrangement	**FA**	• Upper/lower blade bearings, seals • Bearing load and L10 life • Lubrication path from oil system	NR467 Sec.15, [5.3]
5	Kinematic Control Gear (Steering Mechanism)	**FA**	• Control-centre / eccentric arrangement • Control rods, links, crank geometry • Servo actuator interface • Strength of linkage members and pins • Fail-safe / neutral-thrust behaviour	NR467 Sec.15, [5.4]; Sec.14
6	Rotor Casing & Main Bevel Gear	**FA**	• Rotor casing structure and thrust path • Input bevel gear (horizontal drive → vertical rotor) • Gear tooth calculations (ISO 6336) • Main thrust bearing design and life • Lubrication and oil monitoring	NR467 Sec.15, [5.5]

Continued on next page

Table 5.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
7	Bottom Cover / Guard Plate Structure	**FA**	• Guard plate below blade circle • Structural strength (grounding/impact) • Attachment to unit and hull • Detachment / fail-safe considerations	NR467 Sec.15, [5.6]; Part B, Ch.5
8	Hull Seat, Foundation & Well Structure	**FA**	• Foundation ring and bolting • Local hull reinforcement / FEA • Watertight integrity of installation well • Load transfer (thrust, steering moment)	NR467 Sec.15, [5.6]; Part B, Ch.5
9	Seal System (Rotor & Blade Shafts)	**FA**	• Seal type, arrangement and material • Leakage monitoring / oil-in-water prevention • EAL fluid if CLEANSHIP (see Ch.13)	NR467 Sec.15, [5.7]

5.3 VSP Drive, Servo Control & Steering Function

Table 5.2: Voith-Schneider Propeller — Drive & Control

#	Drawing/Document	Category	Content Requirements	Reference
10	Prime Mover Interface	**FA**	• Diesel engine or electric motor input (see Ch.7) • Input coupling / clutch / cardan shaft • Torsional vibration calculation of the complete drive train	NR467 Sec.15; Sec.7

Continued on next page

Table 5.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
11	Servo / Hydraulic Control Unit	**FA**	• Control (steering) hydraulic schematic • Two independent power/servo systems where used for steering (redundancy) • Pressure ratings, relief, accumulators • See Chapter 10	NR467 Sec.14; Ch.10
12	Steering Control System & Bridge Console	**FA**	• Control transmitter (tiller/joystick) to control centre • Thrust magnitude & direction indication • Follow-up / non-follow-up control • Alarms, emergency stop, mode changeover	NR467 Sec.14; Part C, Ch.4
13	Steering Function Compliance Package	**FA**	• FMEA of steering function • Single-failure / duplication analysis • Response time, hard-over to hard-over • SOLAS II-1 Reg.29 demonstration (see Ch.8)	NR467 Sec.14; SOLAS II-1/29

Surveyor note (VSP): A VSP is simultaneously a propeller, a gear unit *and* a steering gear. Do not approve the mechanical package in isolation — confirm that the steering-function documentation of Chapter 8 (redundancy, control stations, alarms, response time, SOLAS compliance) has also been submitted. The blade guard/bottom plate and its hull attachment are frequently under-documented; require structural substantiation against grounding loads.

Chapter 6

Other Thruster Types (Rim-Driven, Pump-Jet, Gill/Jet, Steering Grid)

This chapter addresses less-common propulsor arrangements recognised by BV. Each follows the **general submissions of Chapter 1** plus the requirements of the *nearest analogous conventional type*, with the additional type-specific items below. Confirm the intended notation and service (propulsion, take-me-home, DP, steering) before fixing the package.

6.1 Rim-Driven Thrusters (Integrated Motor Propulsors)

Rim-driven / hubless thrusters integrate a permanent-magnet (or induction) motor whose rotor forms the propeller-blade tip ring inside the duct. There is no mechanical shaft line or reduction gear; excitation and speed are set by a frequency converter. Assessment shifts from gearing toward **electrical machine, converter and duct-bearing** substantiation.

Table 6.1: Rim-Driven Thruster — Type-Specific Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Integrated Motor & Duct Assembly	**FA**	• Motor type (PM / induction), pole/winding data • Rotor rim / blade-tip ring construction • Duct structure and stator support • Cooling (seawater / freshwater path)	NR467 Sec.15; Pt C, Ch.2; IEC 60092-301
2	Frequency Converter / Drive Package	**FA**	• Converter rating and topology • Harmonic / power-quality analysis • Protection, cooling, EMC • See Chapter 7 (electric drive)	NR467 Pt C, Ch.2; IEC 60092-101

Continued on next page

Table 6.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
3	Bearing & Seal Arrangement	**FA**	• Water-lubricated / magnetic bearing design • Bearing load and wear-life basis • Gap / air-gap control, seal design	NR467 Sec.15, [6.1]
4	Blade & Rim Strength Calculations	**FA**	• Blade + rim stress (hydro + magnetic + centrifugal) • Fatigue and corrosion allowance • Cavitation assessment	NR467 Sec.15, [6.1]; Sec.7

6.2 Pump-Jet & Ducted Rotor-Stator Propulsors

Pump-jet units (ducted rotor with stator vanes, often for low-signature or shallow-draught craft) are treated as a ducted propeller + shaft line. Submit the **duct/stator structure and strength, rotor blade design and strength, shaft line (Chapter 9), bearings, seals** and the relevant drive package. Where the unit also steers (deflector/vectoring), add the steering function package of Chapter 8.

6.3 Gill / Hydro-Jet Transverse Thrusters (Pump-Type)

Pump-type transverse thrusters draw water through a hull inlet and discharge through athwartship outlets, providing side thrust without an open tunnel. Submit as for a tunnel thruster (Chapter 2) but replace the tunnel structure items with: **inlet grating and duct, internal pump/impeller design and strength, discharge outlet and directional-valve arrangement, hull penetration and watertight integrity**, plus the pump drive package.

6.4 Steering Grid / Steering-Nozzle & Rudder-Propeller Variants

Fixed-nozzle (Kort-type) units with a steering grid or a steerable nozzle behave as combined propulsion + steering devices. Submit the **nozzle/grid structure and strength, propeller in nozzle, shaft line, steering actuator and control**, and the full steering-gear compliance package of Chapter 8 (SOLAS II-1 Reg.29 where the device is the ship's means of steering).
Surveyor note (other types): For any novel or hybrid propulsor, apply the *function-based* approach — decompose the unit into propeller/impeller, structure, shaft line, gear (if any), drive (electric/diesel/hydraulic), bearings/seals and control — and require the corresponding chapter's submissions for each function. Novel technology may additionally require an Approval-in-Principle (AiP) or risk-based/alternative-design assessment.

Chapter 7

Prime Mover Considerations — Diesel-Engine vs. Electric-Motor Driven Thrusters

7.1 Purpose

The preceding type chapters split the drive into *electric* or *hydraulic*. This chapter consolidates the **additional package elements driven by the choice of prime mover** so the surveyor can confirm the correct drive-train documentation is present. A thruster may be driven by:

- an **electric motor** (constant- or variable-speed, converter-fed), typical of diesel-electric ships, DP vessels, cruise ships and pods; or
- a **diesel engine** (direct/geared mechanical drive), typical of tugs, OSVs and workboats with inboard engines driving Z-drive azimuth or tunnel thrusters via shafting, clutches and gearboxes; or
- a **hydraulic motor** (covered per type chapter and Chapter 10), common on retractable and smaller units.

7.2 Electric-Motor-Driven Thrusters — Additional Submissions

Table 7.1: Electric-Motor Prime Mover — Additional Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Propulsion/Drive Motor Data Sheet	**FA**	• Type (induction / PM / synchronous) • Rated & overload power, torque-speed curve • Voltage, frequency, insulation & IP class • Cooling method, temperature rise (IEC 60034)	NR467 Pt C, Ch.2; IEC 60092-301

Continued on next page

Table 7.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
2	Frequency Converter (VFD) Specification & Arrangement	**FA**	• Converter topology and rating • Output torque ripple at thruster • Protection, redundancy, by-pass • Cooling and heat load to space	NR467 Pt C, Ch.2
3	Harmonic & Power-Quality Analysis	**FA**	• THD at main switchboard (limits per rules) • Effect on generators/other consumers • Mitigation (multi-pulse, active front end, filters)	NR467 Pt C, Ch.2
4	Supply Transformer & Switch-board Interface	**FA**	• Transformer rating, vector group, cooling • Feeder protection and discrimination • Short-circuit & earth-fault co-ordination	NR467 Pt C, Ch.2; IEC 60092-201
5	Power Cable Sizing & Routing	**FA**	• Current rating, voltage drop, derating • Routing, support, bulkhead penetrations • Segregation (DP redundancy where relevant)	IEC 60092-352 / -352
6	Motor Protection, Monitoring & Cooling	**FA**	• Winding/bearing temperature monitoring • Overload, earth-fault, differential protection • Space heaters, forced cooling, condensation control	NR467 Pt C, Ch.2 / Ch.3
7	Torque-Ripple / Torsional Interaction Note	**FA**	• Converter harmonics vs. shaft-line torsionals • Barred-speed / avoidance ranges if any • Cross-reference shaft TVC (Chapter 9)	NR467 Sec.7

Continued on next page

Table 7.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
8	Integration with PMS / Blackout Recovery	FI	• Load-dependent start/stop, load shedding • Blackout and thruster restart sequence	NR467 Pt C, Ch.3

7.3 Diesel-Engine-Driven Thrusters — Additional Submissions

Where a thruster (typically a Z-drive azimuth on a tug/OSV, or a directly driven tunnel/CP unit) is driven by a diesel engine, the **engine and its auxiliary systems become part of the approvable package** under **NR467 Pt C, Ch 1, Sec 2 (Diesel Engines)** in addition to the thruster under Sec 15. A **torsional vibration calculation (TVC) of the complete engine–clutch–shafting–thruster system is mandatory.**

Table 7.2: Diesel-Engine Prime Mover — Additional Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Diesel Engine Approval / Type Data	FA	• Engine general arrangement & rating (MCR) • Type Approval / unit certification status • Crankshaft & main running-gear data • Safety devices (overspeed, LO low-pressure shutdown)	NR467 Pt C, Ch.1, Sec.2
2	Torsional Vibration Calculation (TVC)	FA	• Complete engine–coupling–gear–thruster model • Barred/critical speed ranges • Stresses in shafts, coupling, gear at criticals • Clutch-in/out and misfiring conditions	NR467 Sec.2; Sec.7
3	Clutch & Flexible Coupling	FA	• Clutch type, torque & engagement control • Flexible coupling stiffness/damping data • Alignment and vibration decoupling	NR467 Sec.2; Sec.7

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Table 7.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
4	Engine Foundation & Chocking	**FA**	• Seating structure and holding-down bolts • Resilient mounts / chocking arrangement • Alignment provisions and thrust transmission	NR467 Sec.2; Part B, Ch.5
5	Fuel Oil System	**FA**	• FO service/settling tanks, transfer, filtration • Piping, valves, quick-closing valves • Leakage containment, high-pressure line shielding	NR467 Pt C, Ch.1 (Piping)
6	Lubricating Oil System	**FA**	• LO pumps, cooler, filters, sump • Pressure/temperature monitoring & alarms	NR467 Pt C, Ch.1
7	Cooling Water System	**FA**	• Jacket & charge-air cooling circuits • Sea-water / central cooling, pumps, expansion	NR467 Pt C, Ch.1
8	Starting & Control Air / Starting System	**FA**	• Air receivers, compressors, capacity for starts • Or electric-start battery capacity	NR467 Pt C, Ch.1
9	Exhaust Gas System	**FA**	• Piping, expansion, lagging, silencer • Back-pressure, boundary insulation (fire) • SCR/EGR arrangement if fitted	NR467 Pt C, Ch.1; SOLAS II-2

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Table 7.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
10	Engine Control, Safety & Monitoring	**FA**	• Governor / speed control, overspeed trip • Alarms & safeguards, local/remote control • Integration with thruster & bridge controls	NR467 Pt C, Ch.1 / Ch.3
11	Emissions / NOx Documentation	**FI**	• EIAPP certificate & NOx Technical File • MARPOL Annex VI Tier compliance	MARPOL Annex VI
12	Engine-Room Ventilation & Air Supply	**FI**	• Combustion air and space ventilation • Heat balance / temperature rise	NR467 Pt C, Ch.1

Surveyor note (prime mover): The single most common gap for diesel-driven Z-drive tugs is a **missing or incomplete TVC for the whole train** (engine + clutch + cardan/shaft + azimuth gears + propeller). For electric drives, the recurring gaps are the **harmonic/power-quality study** and **cable-sizing/segregation**. Hybrid PTI/PTO arrangements must document *every* driving path and the mode-changeover logic. Cross-reference the shaft-line torsional/alignment package in Chapter 9.

Chapter 8

Thrusters as Part of Steering Gear

8.1 Applicability

CRITICAL: When thrusters (azimuth, podded, or water-jets) are used as the PRIMARY or AUXILIARY steering system, they MUST comply with ALL steering gear requirements in addition to thruster requirements.

This applies to:

- Azimuth thrusters used for steering
- Podded propulsion units providing steering
- Water-jet systems providing steering
- Any thruster configuration that replaces or supplements rudder steering

8.2 Additional Steering Gear Requirements

All applicable thruster drawings from Chapters 2-4 apply, PLUS the following steering-specific submissions:

8.2.1 Steering System Design

Table 8.1: Steering Gear - System Design

#	Drawing/Document	Category	Content Requirements	Reference
1	Steering System General Arrangement	**FA**	• Complete steering system layout • Main and auxiliary steering provisions • Control stations • Power supply arrangements • Emergency steering provisions	NR467 Sec.14, [1]

Continued on next page

Table 8.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
2	Steering Gear Capacity Calculations	**FA**	• Maximum steering torque • Time to move from 35° port to 35° starboard • Time to move from 35° to 30° on opposite side • Compliance with SOLAS requirements • Redundancy analysis	NR467 Sec.14, [2]; SOLAS Ch.II-1, Reg.29
3	Steering Control System	**FA**	• Control system architecture • Main control station (bridge) • Auxiliary control provisions • Emergency control • Autopilot integration • Manual override provisions	NR467 Sec.14, [3]

8.2.2 Redundancy & Reliability

Table 8.2: Steering Gear - Redundancy Requirements

#	Drawing/Document	Category	Content Requirements	Reference
4	Main and Auxiliary Steering Systems	**FA**	• Main steering arrangement • Auxiliary steering arrangement • Independence of systems • Changeover provisions • Simultaneous operation capability	NR467 Sec.14, [4]; SOLAS Ch.II-1, Reg.29
5	Failure Mode and Effects Analysis (FMEA)	**FA**	• All potential failure modes • Effects on steering capability • Single failure tolerance • Redundancy verification • Mitigation measures	NR467 Sec.14, [5]

Continued on next page

Table 8.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
6	Reliability Analysis	**FA**	• System reliability calculations • Component failure rates • Mean time between failures (MTBF) • Availability analysis • Compliance with steering gear standards	NR467 Sec.14, [5]

8.2.3 Power Supply

Table 8.3: Steering Gear - Power Supply

#	Drawing/Document	Category	Content Requirements	Reference
7	Steering Gear Power Supply	**FA**	• Main power supply arrangement • Emergency power supply • Independent power sources for main/auxiliary • Switchboard arrangements • Circuit protection	NR467 Sec.14, [6]; SOLAS Ch.II-1, Reg.29
8	Power Supply Calculations	**FA**	• Load analysis • Cable sizing • Protection device coordination • Voltage drop calculations • Short circuit analysis	NR467 Sec.14, [6]; IEC 60092-352

8.2.4 Control Stations

Table 8.4: Steering Gear - Control Stations

#	Drawing/Document	Category	Content Requirements	Reference
9	Bridge Control Console	**FA**	• Main steering control layout • Steering wheel or joystick • Rudder angle indicator (or thruster angle) • Rate of turn indicator • Alarm and monitoring displays • Autopilot controls	NR467 Sec.14, [7]

Continued on next page

Table 8.4 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
10	Emergency Steering Station	**FA**	• Emergency control location • Control equipment • Communication with bridge • Indication and monitoring • Access provisions	NR467 Sec.14, [8]; SOLAS Ch.II-1, Reg.29
11	Local Control Station	**FA**	• Local control panel (if applicable) • Control mode selection • Indication and monitoring • Interlock with other control stations	NR467 Sec.14, [9]

8.2.5 Alarms and Monitoring

Table 8.5: Steering Gear - Alarms & Monitoring

#	Drawing/Document	Category	Content Requirements	Reference
12	Alarm System Design	**FA**	• Power supply failure alarms • Control system failure alarms • Actuator failure alarms • Position deviation alarms • Alarm display on bridge	NR467 Sec.14, [10]
13	Monitoring System	**FA**	• Position monitoring • Rate monitoring • Torque/thrust monitoring • Temperature monitoring • Data logging	NR467 Sec.14, [10]

8.2.6 Hydraulic Systems (if applicable)

Table 8.6: Steering Gear - Hydraulic System

#	Drawing/Document	Category	Content Requirements	Reference
14	Steering Gear Hydraulic System	**FA**	• Main hydraulic system • Auxiliary hydraulic system • Independence of systems • Accumulator provisions • See Chapter 10 for detailed requirements	NR467 Sec.14, [11]; Ch.10

8.2.7 Testing & Commissioning

Table 8.7: Steering Gear - Testing Requirements

#	Drawing/Document	Category	Content Requirements	Reference
15	Steering Gear Testing Procedures	**FA**	• Factory acceptance tests • Installation tests • Sea trial procedures • Performance verification • Redundancy testing • Emergency steering tests	NR467 Sec.14, [12]; SOLAS Ch.II-1, Reg.29
16	Steering Gear Test Records	**FI**	• FAT reports • Installation test reports • Sea trial results • Performance verification • As-built documentation	NR467 Sec.14, [12]

8.2.8 Operating Manuals

Table 8.8: Steering Gear - Operating Manuals

#	Drawing/Document	Category	Content Requirements	Reference
17	Steering Gear Operating Manual	**FI**	• Normal operating procedures • Changeover procedures (main/auxiliary) • Emergency procedures • Troubleshooting guide • Safety precautions	NR467 Sec.14, [13]

Continued on next page

Table 8.8 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
18	Maintenance Manual	**FI**	• Maintenance schedule • Inspection procedures • Lubrication requirements • Spare parts list • Overhaul procedures	NR467 Sec.14, [13]

8.2.9 SOLAS Compliance Documentation

Table 8.9: Steering Gear - SOLAS Compliance

#	Drawing/Document	Category	Content Requirements	Reference
19	SOLAS Compliance Statement	**FA**	• Demonstration of compliance with SOLAS Ch.II-1, Reg.29 • Steering gear capacity verification • Redundancy verification • Emergency steering capability • Testing and commissioning compliance	NR467 Sec.14, [14]; SOLAS Ch.II-1, Reg.29

Chapter 9

Thrusters as Part of Main Shafting

9.1 Applicability

When thruster shafts (propeller shafts, intermediate shafts, vertical shafts in azimuth thrusters, or water-jet drive shafts) are considered part of the main propulsion shafting system, they MUST comply with shafting requirements in NR467 Section 7.

This typically applies to:

- Azimuth thruster vertical and horizontal shafts
- Podded propulsion shafts
- Water-jet drive shafts
- Tunnel thruster propeller shafts (for high-power units)
- Any shaft transmitting significant propulsion power

9.2 Overlap with Thruster Requirements

NOTE: Many shaft-related drawings and calculations are already covered in Chapters 2-4. This chapter identifies ADDITIONAL requirements or ENHANCED detail levels required when shafts are classified as "main shafting."

9.2.1 Shaft Design Drawings

Table 9.1: Main Shafting - Design Drawings

#	Drawing/Document	Category	Content Requirements	Reference
1	Shaft Line General Arrangement	**FA**	• Complete shaft line layout • All shaft sections identified • Bearing locations • Coupling locations • Seal locations • Coordinate system	NR467 Sec.7, [1]
2	Shaft Detail Drawings	**FA**	• Dimensional drawings of each shaft section • Diameter changes and transitions • Keyway details • Thread details • Surface finish requirements • Material specifications • Heat treatment requirements	NR467 Sec.7, [2]
3	Shaft End Connections	**FA**	• Flange designs • Coupling designs • Bolt specifications • Fit tolerances • Assembly procedures	NR467 Sec.7, [3]

9.2.2 Shaft Strength Calculations

Table 9.2: Main Shafting - Strength Calculations

#	Drawing/Document	Category	Content Requirements	Reference
4	Shaft Diameter Calculations	**FA**	• Minimum diameter calculations per BV formula • Torsional strength verification • Bending strength verification • Combined stress analysis • Safety factors applied	NR467 Sec.7, [4]

Continued on next page

Table 9.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
5	Shaft Stress Analysis	**FA**	• Detailed stress calculations at critical sections • Stress concentration factors • Keyway effects • Thread effects • FEA results (if applicable)	NR467 Sec.7, [4]
6	Fatigue Analysis	**FA**	• Fatigue life calculations • S-N curves for materials • Stress range analysis • Cumulative damage assessment • Compliance with fatigue criteria	NR467 Sec.7, [5]
7	Deflection Calculations	**FA**	• Shaft deflection under load • Bearing alignment effects • Seal clearance verification • Acceptable deflection limits	NR467 Sec.7, [6]

9.2.3 Critical Speed Analysis

Table 9.3: Main Shafting - Critical Speed Analysis

#	Drawing/Document	Category	Content Requirements	Reference
8	Torsional Vibration Analysis	**FA**	• Natural frequencies calculation • Forced vibration analysis • Resonance identification • Barred speed ranges • Damping analysis • Compliance with acceptable vibration levels	NR467 Sec.7, [7]
9	Lateral Vibration Analysis	**FA**	• Critical speed calculations • Campbell diagram • Operating speed verification • Margin to critical speeds • Bearing stiffness effects	NR467 Sec.7, [7]

Continued on next page

Table 9.3 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
10	Whirling Analysis	**FA**	• Whirling speed calculations • Forward and backward whirl • Operating range verification • Stability analysis	NR467 Sec.7, [7]

9.2.4 Bearing Design

Table 9.4: Main Shafting - Bearing Design

#	Drawing/Document	Category	Content Requirements	Reference
11	Bearing Arrangement Drawing	**FA**	• Bearing types and locations • Bearing housing design • Lubrication system • Seal arrangements • Cooling provisions • Mounting details	NR467 Sec.7, [8]
12	Bearing Load Calculations	**FA**	• Radial loads • Axial loads (thrust bearings) • Moment loads • Load distribution among bearings • Dynamic load factors	NR467 Sec.7, [8]
13	Bearing Life Calculations	**FA**	• L10 life calculations • Operating conditions • Load and speed effects • Lubrication effects • Minimum acceptable life	NR467 Sec.7, [8]
14	Bearing Lubrication System	**FA**	• Lubrication method (oil bath, forced circulation, grease) • Oil type and properties • Oil flow rate (forced systems) • Cooling arrangements • Filtration system • Oil level and temperature monitoring	NR467 Sec.7, [9]

9.2.5 Couplings

Table 9.5: Main Shafting - Couplings

#	Drawing/Document	Category	Content Requirements	Reference
15	Coupling Design Drawing	**FA**	• Coupling type (rigid, flexible) • Dimensions and material • Bolt specifications • Fit tolerances • Assembly procedure	NR467 Sec.7, [10]
16	Coupling Strength Calculations	**FA**	• Torque capacity • Bolt strength verification • Flange strength • Fit stress analysis • Fatigue assessment	NR467 Sec.7, [10]
17	Flexible Coupling Characteristics	**FA**	• Torsional stiffness • Damping properties • Misalignment capacity • Effect on torsional vibration	NR467 Sec.7, [10]

9.2.6 Seals

Table 9.6: Main Shafting - Seals

#	Drawing/Document	Category	Content Requirements	Reference
18	Seal System Design	**FA**	• Seal type and material • Seal arrangement • Pressure rating • Leakage monitoring provisions • Emergency sealing (if applicable)	NR467 Sec.7, [11]
19	Seal Performance Data	**FA**	• Operating pressure and temperature • Shaft speed limitations • Expected leakage rates • Service life • Maintenance requirements	NR467 Sec.7, [11]

9.2.7 Alignment

Table 9.7: Main Shafting - Alignment

#	Drawing/Document	Category	Content Requirements	Reference
20	Shaft Alignment Calculations	**FA**	• Alignment procedure • Bearing reactions • Shaft sag calculations • Thermal growth considerations • Acceptable alignment tolerances	NR467 Sec.7, [12]
21	Alignment Procedure	**FI**	• Step-by-step alignment procedure • Measurement methods • Adjustment provisions • Verification methods • Documentation requirements	NR467 Sec.7, [12]

9.2.8 Material Specifications

Table 9.8: Main Shafting - Material Specifications

#	Drawing/Document	Category	Content Requirements	Reference
22	Shaft Material Specifications	**FA**	• Material grades (steel, stainless steel, etc.) • Mechanical properties • Heat treatment requirements • Surface treatment (if applicable) • Corrosion protection • Compliance with BV approved materials	NR467 Sec.7, [13]; Part B, Ch.3
23	Material Certificates	**FA**	• Material test certificates (during construction) • Chemical composition • Mechanical properties • Heat treatment records • Traceability	NR467 Part B, Ch.3

9.2.9 Non-Destructive Testing (NDT)

Table 9.9: Main Shafting - NDT Requirements

#	Drawing/Document	Category	Content Requirements	Reference
24	NDT Plan	**FA**	• NDT methods (UT, MT, PT, RT) • Extent of testing • Acceptance criteria • Inspector qualifications • Testing stages	NR467 Sec.7, [14]; Part B, Ch.4
25	NDT Reports	**FI**	• NDT test reports (during construction) • Test results • Defect identification and disposition • Inspector certification	NR467 Part B, Ch.4

9.2.10 Testing & Commissioning

Table 9.10: Main Shafting - Testing & Commissioning

#	Drawing/Document	Category	Content Requirements	Reference
26	Shaft Testing Procedures	**FA**	• Dimensional verification • Material testing • NDT • Balance testing (if applicable) • Assembly testing • Acceptance criteria	NR467 Sec.7, [15]
27	Vibration Monitoring Plan	**FA**	• Vibration measurement locations • Measurement methods • Acceptance criteria • Monitoring during sea trials • Long-term monitoring provisions	NR467 Sec.7, [16]
28	Test Reports	**FI**	• Factory test reports • Installation test reports • Sea trial vibration measurements • Alignment verification • As-built documentation	NR467 Sec.7, [15]

Chapter 10

Hydraulic Systems & Pitch Control Mechanisms

10.1 Applicability

Hydraulic systems are used in thrusters for:

- Hydraulic motor drive (alternative to electric motor)
- Retractable thruster lifting/lowering mechanism
- Controllable pitch propeller pitch control
- Azimuth slewing drive (if hydraulic)
- Water-jet steering and reversing actuation

10.2 Hydraulic Power Unit (HPU)

Table 10.1: Hydraulic Systems - HPU

#	Drawing/Document	Category	Content Requirements	Reference
1	HPU General Arrangement	**FA**	• Complete HPU layout • Pump arrangement • Motor/drive arrangement • Reservoir design • Valve manifold • Accumulator location • Filtration system • Cooling system • Mounting arrangement	NR467 Sec.15

Continued on next page

Table 10.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
2	Hydraulic Pump Specification	**FA**	• Pump type (gear, piston, vane) • Displacement • Rated pressure and flow • Drive motor specifications • Efficiency curves • Manufacturer and model	NR467 Sec.15
3	Hydraulic Reservoir Design	**FA**	• Reservoir capacity • Material and construction • Baffle arrangement • Oil level indication • Breather and filler • Drain provisions • Heating elements (if applicable)	NR467 Sec.15
4	HPU Calculations	**FA**	• Flow requirements • Pressure requirements • Pump sizing verification • Reservoir sizing • Heat generation and cooling capacity • Accumulator sizing (if applicable)	NR467 Sec.15

10.3 Hydraulic Circuit Design

Table 10.2: Hydraulic Systems - Circuit Design

#	Drawing/Document	Category	Content Requirements	Reference
5	Hydraulic System Schematic	**FA**	• Complete hydraulic circuit • All components identified with symbols • Pressure ratings • Flow ratings • Control logic • Safety devices • Monitoring points	NR467 Sec.15

Continued on next page

Table 10.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
6	Piping and Instrumentation Diagram (P&ID)	**FA**	• Detailed piping layout • Pipe sizes and materials • Valve locations and specifications • Instrument locations • Pressure and temperature monitoring • Drain and vent points	NR467 Sec.15
7	Hydraulic Piping Arrangement	**FA**	• 3D piping layout (or isometric) • Pipe routing • Pipe supports • Flexibility analysis • Penetrations through bulkheads • Accessibility for maintenance	NR467 Sec.15

10.4 Hydraulic Components

Table 10.3: Hydraulic Systems - Components

#	Drawing/Document	Category	Content Requirements	Reference
8	Hydraulic Motor Specification	**FA**	• Motor type and displacement • Rated pressure and speed • Torque and power curves • Efficiency • Mounting details • Manufacturer and model	NR467 Sec.15
9	Hydraulic Cylinder Specification	**FA**	• Cylinder bore and stroke • Operating pressure • Rod diameter and material • Seal specifications • Mounting arrangement • Position sensors • Manufacturer and model	NR467 Sec.15

Continued on next page

Table 10.3 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
10	Control Valve Specifications	**FA**	• Valve types (directional, pressure, flow) • Valve ratings • Actuation method (solenoid, manual, pilot) • Response time • Manufacturer and model	NR467 Sec.15
11	Pressure Relief Valve	**FA**	• Set pressure • Flow capacity • Type (direct-acting, pilot-operated) • Mounting location • Manufacturer and model	NR467 Sec.15
12	Accumulator Specification	**FA**	• Accumulator type (bladder, piston, diaphragm) • Volume • Pre-charge pressure • Operating pressure range • Gas type (nitrogen) • Mounting arrangement • Safety provisions	NR467 Sec.15
13	Accumulator Sizing Calculations	**FA**	• Volume requirements • Pre-charge pressure calculation • Energy storage capacity • Response time analysis	NR467 Sec.15

10.5 Filtration System

Table 10.4: Hydraulic Systems - Filtration

#	Drawing/Document	Category	Content Requirements	Reference
14	Filtration System Design	**FA**	• Filter types and locations • Filtration rating (micron) • Flow capacity • Pressure drop • Filter change indicators • Bypass provisions	NR467 Sec.15
15	Filter Specifications	**FA**	• Filter element type • Filtration rating • Collapse pressure • Manufacturer and model • Replacement schedule	NR467 Sec.15

10.6 Cooling System

Table 10.5: Hydraulic Systems - Cooling

#	Drawing/Document	Category	Content Requirements	Reference
16	Hydraulic Oil Cooling System	**FA**	• Cooling method (air, water) • Heat exchanger specifications • Cooling capacity • Temperature control • Monitoring provisions	NR467 Sec.15
17	Thermal Analysis	**FA**	• Heat generation calculations • Heat dissipation calculations • Temperature rise predictions • Cooling capacity verification • Operating temperature range	NR467 Sec.15

10.7 Hydraulic Fluid

Table 10.6: Hydraulic Systems - Fluid Specification

#	Drawing/Document	Category	Content Requirements	Reference
18	Hydraulic Fluid Specification	**FA**	• Fluid type (mineral oil, synthetic, biodegradable) • Viscosity grade • Operating temperature range • Fire resistance properties (if applicable) • Compatibility with system materials • Approved manufacturer and grade	NR467 Sec.15

10.8 Piping Design

Table 10.7: Hydraulic Systems - Piping

#	Drawing/Document	Category	Content Requirements	Reference
19	Pipe Sizing Calculations	**FA**	• Flow velocity calculations • Pressure drop calculations • Pipe diameter selection • Compliance with velocity limits	NR467 Sec.15
20	Pipe Material Specifications	**FA**	• Pipe material (steel, stainless steel, copper-nickel) • Wall thickness • Pressure rating • Corrosion protection • Compliance with BV approved materials	NR467 Sec.15; Part B, Ch.3
21	Pipe Support Design	**FA**	• Support locations and types • Support spacing • Vibration isolation • Thermal expansion accommodation • Load calculations	NR467 Sec.15

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Table 10.7 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
22	Piping Stress Analysis	**FA**	• Pressure stress • Thermal stress • Vibration analysis • Support reaction forces • Compliance with stress limits	NR467 Sec.15

10.9 Hose Assemblies

Table 10.8: Hydraulic Systems - Hoses

#	Drawing/Document	Category	Content Requirements	Reference
23	Hydraulic Hose Specifications	**FA**	• Hose type and construction • Pressure rating • Temperature rating • End fittings • Length and routing • Fire resistance (if required) • Manufacturer and model	NR467 Sec.15
24	Hose Installation Drawing	**FA**	• Hose routing • Bend radius verification • Support and protection • Accessibility for inspection/replacement	NR467 Sec.15

10.10 Safety Devices

Table 10.9: Hydraulic Systems - Safety Devices

#	Drawing/Document	Category	Content Requirements	Reference
25	Pressure Relief System	**FA**	• Relief valve locations • Set pressures • Discharge routing • Capacity verification	NR467 Sec.15

Continued on next page

Table 10.9 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
26	Emergency Shutdown System	**FA**	• Emergency stop provisions • Valve actuation • System depressurization • Safe state definition	NR467 Sec.15
27	Leak Detection System	**FA**	• Leak detection method • Alarm provisions • Automatic shutdown (if required)	NR467 Sec.15

10.11 Control System

Table 10.10: Hydraulic Systems - Control

#	Drawing/Document	Category	Content Requirements	Reference
28	Hydraulic Control System	**FA**	• Control architecture • Manual/automatic control modes • Valve actuation signals • Feedback systems • Safety interlocks	NR467 Sec.15; Part C, Ch.4
29	Control Panel Layout	**FA**	• Panel arrangement • Control devices • Indication and monitoring • Alarm displays • Emergency stop	NR467 Sec.15

10.12 Monitoring & Instrumentation

Table 10.11: Hydraulic Systems - Monitoring

#	Drawing/Document	Category	Content Requirements	Reference
30	Instrumentation List	**FA**	• Pressure gauges/transmitters • Temperature sensors • Flow meters (if applicable) • Level indicators • Position sensors • Filter condition indicators • Specifications for each instrument	NR467 Sec.15
31	Alarm and Monitoring System	**FA**	• Parameters monitored • Alarm setpoints • Alarm display location • Data logging (if applicable)	NR467 Sec.15

10.13 Controllable Pitch Mechanism (CP Propellers)

Table 10.12: Hydraulic Systems - CP Mechanism

#	Drawing/Document	Category	Content Requirements	Reference
32	CP Mechanism Assembly	**FA**	• Hub internal mechanism • Pitch control rod/linkage • Hydraulic cylinder (in hub or external) • Blade retention system • Bearing arrangement • Seal design	NR467 Sec.15, [3.4]
33	CP Mechanism Calculations	**FA**	• Pitch control forces • Hydraulic cylinder sizing • Linkage strength • Blade retention strength • Seal pressure rating	NR467 Sec.15, [3.4]
34	Pitch Control Hydraulic System	**FA**	• Hydraulic supply to hub • Rotary joint/slip ring design • Control valve system • Pitch feedback system • Emergency pitch setting	NR467 Sec.15, [3.4]

Continued on next page

Table 10.12 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
35	Pitch Feedback System	**FA**	• Position sensor type and location • Signal transmission (rotary joint) • Accuracy and resolution • Redundancy (if required)	NR467 Sec.15, [3.4]

10.14 Testing & Commissioning

Table 10.13: Hydraulic Systems - Testing

#	Drawing/Document	Category	Content Requirements	Reference
36	Hydraulic System Test Procedures	**FA**	• Pressure testing • Functional testing • Leak testing • Performance verification • Safety device testing • Acceptance criteria	NR467 Sec.15
37	Hydraulic System Test Reports	**FI**	• Pressure test results • Functional test results • Leak test results • Performance data • As-built documentation	NR467 Sec.15

10.15 Operating & Maintenance Manuals

Table 10.14: Hydraulic Systems - Manuals

#	Drawing/Document	Category	Content Requirements	Reference
38	Hydraulic System Operating Manual	**FI**	• System description • Operating procedures • Normal operation • Emergency procedures • Troubleshooting guide • Safety precautions	NR467 Sec.15

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Table 10.14 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
39	Hydraulic System Maintenance Manual	**FI**	• Maintenance schedule • Inspection procedures • Fluid sampling and analysis • Filter replacement • Component overhaul • Spare parts list	NR467 Sec.15

Chapter 11

Dynamic Positioning (DP) Thrusters - Additional Requirements

11.1 Applicability

Thrusters that are part of a Dynamic Positioning (DP) system must comply with ALL applicable requirements from Chapters 2-10, PLUS the additional DP-specific requirements in this chapter.

DP systems are classified into three classes:

- **DP-1**: Basic DP capability, no redundancy required
- **DP-2**: Redundant systems, single failure tolerance
- **DP-3**: Enhanced redundancy with fire/flood separation

The DP-specific submissions below are *in addition to* the full type package for each thruster used in the DP system (structure, propeller, shaft, gear, drive, control). They focus on redundancy, power segregation, control and failure analysis, and are assessed principally under **NR216** and **NR467 Part E**.

Key DP-Specific Documents Required (FA):

1. DP System General Arrangement
2. Thruster Configuration Drawing
3. DP Capability Plot
4. FMEA Documentation (comprehensive)
5. Redundancy Concept (DP-2/DP-3)
6. Power Supply Segregation
7. Fire Zone Arrangement (DP-3)
8. Watertight Subdivision (DP-3)
9. DP Control System Architecture
10. Thruster Allocation Logic
11. Position Reference System Configuration

12. Environmental Sensor Configuration
13. Power Management System (PMS) Configuration
14. UPS Configuration
15. DP FMEA Proving Trials Plan
16. DP Annual Trials Plan

Key DP-Specific Documents Required (FI):

1. DP Operations Manual
2. DP Maintenance Manual
3. DP Trials Reports
4. DP Training Requirements

11.2 Reference

All DP requirements per **NR216 - Rules for Dynamic Positioning Systems**

Chapter 12

Ice Class & Cold-Climate (Winterisation) Requirements

12.1 Applicability

When a vessel carries a BV ice-class or polar notation, thrusters and their drive trains must be strengthened and winterised in addition to all base requirements of the preceding chapters. Two families apply:

- **Baltic / Finnish-Swedish Ice Classes** — notations **ICE CLASS IA SUPER, IA, IB, IC** (and **ID**), based on the Finnish-Swedish Ice Class Rules (FSICR / TRAFICOM), adopted in **NR467 Part E**. Intended for first-year Baltic-type ice with icebreaker assistance.
- **Polar Classes** — notations **POLAR CLASS PC1–PC7** (IACS Polar Class / IMO Polar Code), addressed in **NR467 Part E** and Rule Note **NR527**. Machinery strengthening follows **IACS UR I3**.

Cold-climate service without ice loads is covered by winterisation notations such as **COLD** (e.g. `COLD(H`t_{DH}`, E`t_{DE}`)`), addressing design ambient temperatures rather than ice impact.

Key principle: ice class primarily drives *ice-load strengthening* of the propeller, shaft line, gears and thruster body, plus *winterisation* of exposed and seawater systems. The surveyor overlays these on the type package.

12.2 Ice-Strengthening Submissions

Table 12.1: Ice Class — Additional Thruster Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Ice Class Notation Basis & Design Conditions	**FA**	• Declared notation (IA Super/IA/... or PC1–PC7) • Design ice thickness / ice-load basis • Applicable rule edition and lower design temperature	NR467 Part E; NR527
2	Propeller Ice-Load Strength Calculation	**FA**	• Ice interaction loads (milling, impact) on blades • Blade root and blade tip strength vs. ice bending • Blade material and thickness increase • Open vs. ducted / nozzle ice loads	NR467 Part E; IACS UR I3
3	Shaft Line & Ice Torque Verification	**FA**	• Maximum ice-induced torque and thrust • Propeller shaft, intermediate shaft, couplings • Overload/peak-torque capacity of gears • Blade-loss / one-blade-milling scenarios	NR467 Part E; Sec.7; IACS UR I3
4	Gear & Azimuth/Nozzle Ice-Load Substantiation	**FA**	• Bevel/reduction gears vs. ice peak torque • Azimuth slewing bearing/gear vs. ice-block impact on lower unit • Nozzle/steering-grid ice-impact strength	NR467 Part E; Sec.15
5	Thruster Body / Lower Unit Ice Protection	**FA**	• Lower gearbox/pod shape vs. ice interaction • Ice knife / deflector where required • Local structural strengthening	NR467 Part E

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Table 12.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
6	Tunnel / Inlet Ice & Grid Arrangement	**FA**	• Tunnel thruster grid ice/debris protection • De-icing / anti-blocking provisions • Ice-load on grid bars	NR467 Part E; Sec.15
7	Winterisation of Systems (Hydraulic, Cooling, Seals)	**FA**	• Low-temperature grades of oils/greases/seals • Heating/insulation of exposed hydraulics & sensors • Sea-chest/cooling anti-icing (recirculation) • Materials for low-temperature toughness (Charpy)	NR467 Part E (COLD); Part B, Ch.3
8	Materials Low-Temperature Certification	**FA**	• Impact testing at design temperature • Grade selection for exposed & load-bearing parts	NR467 Part B, Ch.3; Part E
9	Retraction/Deployment in Ice (if retractable)	**FI**	• Operating limits and procedures in ice • Protection of opening/seals from ice ingress	NR467 Part E

Surveyor note (ice): The governing case for ice-classed propulsion is usually the **ice peak torque / blade-milling load** propagating through propeller → shaft → gear → prime mover, *not* the open-water rating. Verify the ice-load calculation explicitly and check that it is carried consistently into the shaft (Chapter 9) and gear strength checks. For azimuth/podded units in ice, confirm lower-unit and slewing-gear substantiation against ice-block impact — a frequent omission. Match the rule edition to the declared notation.

Chapter 13

Additional BV Class Notations Affecting Thruster Packages

13.1 Purpose

Beyond ice class, several BV **additional class notations** and **service notations** impose extra requirements on the thruster package. This chapter summarises the ones most often encountered so the surveyor can confirm the client has submitted the corresponding documents. Requirements here *add to* the type package; they do not replace it.

13.2 Notation Impact Matrix

Table 13.1: Additional Notations — Impact on Thruster Submissions

Notation (family)	Concern	Additional thruster submissions	Reference
DYNAPOS (AM/AT/ATR, SAM/SAT); **DP-1/2/3**	Dynamic positioning	Full DP package — see Chapter 11: FMEA, redundancy, power segregation, capability plot, thruster allocation, proving trials.	NR467 Pt E; NR216
AUT-UMS, AUT-CCS, AUT-PORT	Automated / unattended machinery	Extended alarm & monitoring of thruster (bearing temps, oil, vibration), remote & safe control, fail-safe on power/control loss, integration with alarm system.	NR467 Pt C, Ch.3; Pt E
COMF-NOISE / COMF-VIB (Comfort)	Noise & vibration on board	Structure-borne & airborne noise prediction, vibration levels at thruster excitation (blade-rate), resilient mounting, cavitation-noise control, source data.	NR467 Pt E, Ch.6

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Table 13.1 – continued from previous page

Notation (family)	Concern	Additional thruster submissions	Reference
CLEANSHIP / environmental	Pollution prevention	Environmentally Acceptable Lubricants (EAL) for sterntube/thruster seals & hydraulics, leakage monitoring, oil-to-sea interface documentation.	NR467 Pt E (environmental); VGP/EAL
Redundant Propulsion (RP, RP-1, AVM-. . .)	Propulsion availability	Independence/segregation of propulsion trains, single-failure analysis, take-me-home capability, redundant thruster arrangement.	NR467 Pt E
TUG / ESCORT TUG / AHT (service)	High steering & towing loads	Escort/indirect-towing loads on azimuth & steering system, structural substantiation of unit & foundation, stability interaction, uprated steering duty.	NR467 Pt D
SPS (Special Purpose Ship)	Personnel / mission systems	Higher survivability & redundancy expectations may extend to propulsion/steering arrangement.	NR467 Pt D/E
Electric hybrid / battery	Hybrid power	Documentation of every driving path (PTI/PTO), energy-storage interface, mode changeover, converter data.	NR467 Pt E

Surveyor note (notations): Always cross-check the vessel's *requested class notation string* against the thruster package. A DP or comfort notation in particular pulls in substantial extra documentation (FMEA/proving trials; noise & vibration studies) that clients frequently omit from the initial thruster submission. Where a notation adds a steering-critical duty (escort tug), confirm the steering-gear package (Chapter 8) reflects the *uprated* loads, not just the open-water condition.

Chapter 14

Ship-Type & Service-Notation Specific Requirements

14.1 Purpose

The base thruster package (Chapters 1–6) and the prime-mover, ice and notation overlays (Chapters 7, 12, 13) define *what a thruster needs*. This chapter answers the complementary question the surveyor must also ask: **"given the ship type / service notation, what *extra* do I expect in the thruster package, and which *additional ship rules* apply?"**

BV assigns each ship a **service notation** (NR467 **Part D**) — e.g. `tug`, `supply`, `oil tanker`, `passenger ship` — plus **additional class notations** (Part E). Together these pull in requirements beyond NR467 Sec. 15, and sometimes bring in *separate rule sets* (offshore units, naval ships, high-speed craft). The matrix below is the surveyor's "what else to demand" guide; the subsequent sections expand the cases with the most rule impact.

14.2 Ship-Type → Thruster Expectation Matrix

Table 14.1: Ship Type / Service — Typical Arrangement & Additional Requirements

Ship type / service	Typical thruster arrangement	Additional requirements & notations to expect	Reference
Harbour / escort tug	Azimuth Z-drive or VSP (main propulsion + steering)	Escort/indirect-towing steering & structural loads; uprated slewing duty; steering redundancy; stability interaction; full steering-gear package.	NR467 Pt D (tug/escort tug); Sec.14
Offshore supply / PSV / AHTS	Bow tunnel(s) + retractable + stern azimuth; DP	DYNAPOS/DP (usually DP-2): FMEA, redundancy, power segregation, capability plot; FiFi where fitted; station-keeping.	NR216; Pt D/E
Dredger (TSHD / CSD)	Bow tunnels; azimuth; sometimes DP	DREDGING service; DP for some; robust manoeuvring; abrasion/duty considerations.	NR467 Pt D

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Table 14.1 – continued from previous page

Ship type / service	Typical thruster arrangement	Additional requirements & notations to expect	Reference
Passenger ship / RoPax / ferry	Bow (+ stern) tunnels; azimuth/pods on some	SOLAS steering (II-1/29); **Safe Return to Port** survivability & redundancy for ships ≥120 m / 3 fire zones; COMF.	SOLAS II-1/II-2; Pt E
Cruise ship	Podded propulsors + multiple bow/stern tunnels	SRtP; COMF-NOISE/VIB; propulsion/steering redundancy; sometimes DP.	SOLAS II-2 Reg.21/22; Pt E
Double-ended ferry	Azimuth or VSP at both ends	Steering redundancy *at each end*; control-station changeover; response time both directions.	Sec.14; SOLAS II-1/29
Cable / pipe layer, offshore construction	Retractable + azimuth + tunnels; DP	DYNAPOS DP-2/DP-3: FMEA, fire/flood separation (DP-3), redundancy, proving trials.	NR216; Pt E
Drillship / semi-sub (MODU)	Multiple azimuth thrusters; DP-3	DP-3 station keeping; offshore-unit rules; fire/flood zone separation; extensive FMEA.	NR216; MODU / offshore rules
Oil / chemical tanker	Bow thruster (manoeuvring)	**Hazardous-area / explosion protection** where equipment is in or adjacent to dangerous zones; thruster-room zoning; certified-safe (Ex) equipment.	SOLAS II-2; IEC 60079; Pt C
Gas carrier (LNG/LPG)	Bow thruster; sometimes azimuth	As tanker *plus* IGC Code hazardous-zone requirements; gas-safe equipment.	IGC Code; IEC 60079
Ice-class / polar trader	Ice-strengthened tunnel/azimuth	Ice-load strengthening & winterisation (see Chapter 12).	NR467 Pt E; NR527
Research / naval / patrol	Rim-driven / pods / azimuth	Low underwater-noise & signature; COMF/silent; naval ships follow dedicated naval rules.	Naval rules (e.g. NR483); Pt E
Large yacht	Bow/stern tunnels; pods	Comfort (noise/vibration); yacht-specific rules; retractable/stabiliser interfaces.	Yacht rules; Pt E
Container / bulk / general cargo	Bow tunnel thruster (manoeuvring)	Base Sec. 15 package; typically electric drive; no special service overlay.	NR467 Sec.15; Pt C

14.3 Passenger Ships & Safe Return to Port (SRtP)

For passenger ships to which SOLAS SRtP applies (broadly, ships ≥120 m in length or having three or more main vertical fire zones), the ship must retain, after a defined casualty (fire in one zone or flooding of one watertight compartment), a set of **essential systems — including propulsion and steering**. Where thrusters/pods/steering thrusters form part of the propulsion or steering means, the surveyor must confirm:

- redundancy and physical/functional separation so a single casualty does not disable all propulsion & steering;
- power supply, control and cooling for the retained thruster(s) surviving the casualty (fire zone / flooded compartment);
- consistency with the ship's SRtP systems-availability documentation and casualty-threshold analysis.

This typically *adds* redundancy/segregation drawings and a systems availability analysis to the base thruster package. *(Ref. SOLAS II-2 Reg.21/22; NR467 passenger-ship requirements.)*

14.4 Tankers & Gas Carriers — Hazardous Areas & Explosion Protection

On oil tankers, chemical tankers and gas carriers, the bow-thruster room and any electrical/rotating equipment located in or adjacent to **hazardous (gas-dangerous) zones** must meet explosion-protection requirements. The surveyor should verify:

- hazardous-area classification of the thruster space and boundaries;
- **certified safe-type (Ex) equipment** for motors, sensors and junction boxes where required, with certificates;
- cable entries, penetrations and ventilation preserving zone integrity;
- for gas carriers, compliance with the **IGC Code** zoning in addition to SOLAS II-2.

(Ref. SOLAS II-2; IEC 60079 series; IGC Code; NR467 Pt C electrical in hazardous areas.) A tanker bow thruster is often electric; confirm the motor's location relative to the cargo/hazardous zone *before* accepting standard (non-Ex) equipment.

14.5 Offshore & DP-Intensive Vessels

OSVs, construction, cable/pipe-lay units and MODUs are defined by their DP capability. Beyond the DP package of Chapter 11, expect: multiple independent thruster trains, power-plant segregation, DP-3 fire/flood separation where notated, worst-case-failure design intent, and comprehensive FMEA and proving-trials programmes. Mobile offshore units are additionally governed by **offshore-unit / MODU rules**, which may modify hull, machinery and location (hazardous-area) requirements relative to NR467.

14.6 Tugs & Escort Tugs

For tugs the thruster *is* the main propulsion and steering means (Z-drive or VSP). Escort tugs generate very high **indirect steering and towing loads**; the surveyor must confirm the azimuth/steering system, unit structure and hull foundation are substantiated for these loads (not merely free-running thrust), together with the full steering-gear package (Chapter 8) and stability interaction. *(Ref. NR467 Pt D tug / escort-tug service notations.)*

14.7 Thruster Compartment (All Ship Types)

Independent of ship type, the thruster room / bow-thruster compartment carries its own requirements the surveyor should not overlook: watertight boundaries and subdivision, bilge/drainage, fire protection and boundary insulation, ventilation and (for tankers/gas carriers) hazardous-area integrity, plus safe access and escape. Confirm these are reflected in the general arrangement and hull/piping documentation. *(Ref. NR467 Part B, Ch 5; Part C fire & piping; SOLAS II-2.)*
Surveyor note (ship type): Read the *requested notation string* first — it tells you the ship type, service and additional notations, and therefore which overlays of Chapters 7–13 and which *separate rule sets* (SRtP, hazardous-area/IGC, offshore/MODU, naval, HSC) apply. The most common ship-type gaps are: SRtP redundancy on passenger ships, Ex/hazardous-area equipment on tankers and gas carriers, and DP-3 separation on offshore units.

Chapter 15

Testing & Commissioning Requirements

15.1 Factory Acceptance Tests (FAT)

Required for All Thrusters:

- Dimensional verification
- Material verification
- Non-destructive testing
- Assembly inspection
- Mechanical running test
- Hydraulic system test (if applicable)
- Electric motor test (if applicable)
- Control system test
- Retraction mechanism test (retractable)
- CP mechanism test (CP propellers)
- Azimuth slewing test (azimuth thrusters)

15.2 Installation Tests

On-Site Verification:

- Foundation inspection
- Alignment verification
- Piping inspection
- Electrical installation inspection
- Control system integration test
- Hydraulic system commissioning

15.3 Dock Trials

Pre-Sea Trials:

- Static thrust test
- No-load running test
- Control system functional test
- Retraction/extension test (retractable)
- Azimuth rotation test (azimuth)
- CP pitch control test (CP propellers)

15.4 Sea Trials

Performance Verification:

- Performance verification
- Maneuverability trials
- Vibration measurements
- Noise measurements
- Thermal performance
- Steering gear trials (if steering function)
- DP trials (for DP systems)
- DP FMEA proving trials (for DP systems)

15.5 Documentation & Reporting

Required Reports (FI):

- FAT Report
- Installation Report
- Sea Trial Report
- DP Trials Report (for DP systems)
- As-Built Documentation

Chapter 16

Surveyor Checklists by Thruster Type

16.1 Tunnel Thrusters (Fixed) - Complete Checklist

16.1.1 General Information

☐ Thruster power rating >110 kW (Section 15 applies)

☐ Thruster manufacturer and model

☐ Installation location (bow/stern)

16.1.2 Structural

☐ General Arrangement Drawing (**FA**)

☐ Tunnel Structural Arrangement (**FA**)

☐ Tunnel Structural Calculations (**FA**)

☐ Tunnel End Grating/Grid Design (**FA**)

☐ Hull Penetration Details (**FA**)

16.1.3 Thruster Unit - Mechanical

☐ Thruster Unit General Assembly (**FA**)

☐ Propeller Design Drawing (**FA**)

☐ Propeller Strength Calculations (**FA**)

☐ Propeller Shaft Drawing (**FA**)

☐ Shaft Strength Calculations (**FA**)

☐ Bearing Arrangement Drawing (**FA**)

☐ Bearing Load Calculations (**FA**)

☐ Gearbox Assembly Drawing (**FA**)

☐ Gear Tooth Calculations (**FA**)

☐ Gearbox Lubrication System (**FA**)

16.1.4 Drive System

If Electric:

- ☐ Electric Motor Specification (**FA**)
- ☐ Motor Foundation Drawing (**FA**)
- ☐ Motor-Gearbox Coupling (**FA**)

If Hydraulic:

- ☐ Hydraulic Motor Specification (**FA**)
- ☐ Hydraulic System Schematic (**FA**)
- ☐ See Chapter 10 for complete hydraulic requirements

16.1.5 Control System

- ☐ Control System Block Diagram (**FA**)
- ☐ Control Panel Layout (**FA**)
- ☐ Electrical Single-Line Diagram (**FA**)
- ☐ Control Software Description (**FA**)

16.1.6 Calculations

- ☐ Thrust Calculation (**FA**)
- ☐ Vibration Analysis (**FA**)
- ☐ Noise Level Prediction (**FA**)
- ☐ Thermal Analysis (**FA**)

16.1.7 Materials & Welding

- ☐ Material Specifications (**FA**)
- ☐ Welding Procedures (**FA**)
- ☐ Material Certificates (during construction)
- ☐ NDT Plan (**FA**)

16.1.8 Testing

- ☐ FAT Procedures (**FA**)
- ☐ Installation Test Procedures (**FA**)
- ☐ Sea Trial Procedures (**FA**)
- ☐ Test Reports (**FI** - after testing)

16.1.9 Manuals

- ☐ Operation & Maintenance Manual (**FI**)
- ☐ Spare Parts List (**FI**)
- ☐ Installation Manual (**FI**)

16.2 Retractable Thrusters - Complete Checklist

All items from Section 10.1 (Tunnel Thrusters), PLUS:

16.2.1 Retraction Mechanism

- ☐ Retraction Mechanism Assembly (**FA**)
- ☐ Hydraulic Cylinder Specifications (**FA**)
- ☐ Retraction System Calculations (**FA**)
- ☐ Hull Opening & Closure Design (**FA**)
- ☐ Hull Opening Structural Analysis (**FA**)
- ☐ Watertight Integrity Documentation (**FA**)
- ☐ Locking System Design (**FA**)
- ☐ Position Indication System (**FA**)
- ☐ Emergency Lowering System (**FA**)
- ☐ Hydraulic System for Retraction (**FA**)

16.2.2 Additional Manuals

- ☐ Retraction/Extension Procedures (**FI**)
- ☐ Emergency Recovery Procedures (**FI**)

16.2.3 Additional Testing

- ☐ Retraction mechanism FAT
- ☐ Seal integrity testing
- ☐ Emergency lowering test
- ☐ Dock trials - retraction/extension
- ☐ Sea trials - retraction under way (if applicable)

16.3 Azimuth Thrusters (Fixed Pitch) - Complete Checklist

16.3.1 General & Structure

- ☐ General Arrangement Drawing incl. 360° clearance envelope (**FA**)
- ☐ Azimuth Unit Complete Assembly (**FA**)
- ☐ Hull Penetration & Foundation (**FA**)
- ☐ Hull Structure Calculations / FEA (**FA**)

16.3.2 Propeller & Shaft

- ☐ Propeller Design Drawing (**FA**)
- ☐ Propeller Strength Calculations (**FA**)
- ☐ Vertical Shaft Assembly & Calculations (**FA**)
- ☐ Thrust Bearing Design & Calculations (**FA**)
- ☐ Radial Bearing Arrangement (**FA**)
- ☐ Column Structure Design (**FA**)
- ☐ Seal System Design (**FA**)

16.3.3 Gear & Slewing System

- ☐ Gear Train Assembly (**FA**)
- ☐ Gear Tooth Strength Calculations — ISO 6336 (**FA**)
- ☐ Gear Lubrication System (**FA**)
- ☐ Slewing Bearing Design & Calculations (**FA**)
- ☐ Azimuth Drive Mechanism & Calculations (**FA**)
- ☐ Position Feedback System (**FA**)

16.3.4 Drive System & Prime Mover

- ☐ Electric or hydraulic power-transmission package (**FA**)
- ☐ Slip ring / rotary joint (electric) (**FA**)
- ☐ **Prime-mover add-on** — electric or diesel (Chapter 7)

16.3.5 Control, Calculations, Materials & Testing

- ☐ Control System Architecture & Bridge Console (**FA**)
- ☐ Local Control Station (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Material Specifications, Welding, NDT Plan (**FA**)
- ☐ FAT / dock trial (azimuth slewing) / sea-trial procedures (**FA**)
- ☐ O&M and Installation Manuals (**FI**)

16.3.6 If Steering / DP / Ice / Notations

- ☐ If steering function → Chapter 8 package (SOLAS II-1/29, FMEA)
- ☐ If part of main shaft line → Chapter 9 package
- ☐ If DP → Section 16. (DP add-on below)
- ☐ If ice class → Chapter 12 add-on
- ☐ Overlay applicable notations → Chapter 13

16.4 Azimuth Thrusters (Controllable Pitch) - Complete Checklist

All items from Azimuth FP, PLUS:

- ☐ CP Hub Mechanism Assembly (**FA**)
- ☐ Blade Retention Calculations (**FA**)
- ☐ Pitch Control Mechanism (**FA**)
- ☐ Pitch Control Hydraulic System incl. rotary joint (**FA**) — Chapter 10
- ☐ Pitch feedback & fail-safe position (**FA**)

16.5 Podded Propulsion Units - Complete Checklist

All items from Azimuth (FP or CP), PLUS:

- ☐ Pod-specific structural components & strut (**FA**)
- ☐ Electric motor in pod — specification, cooling, monitoring (**FA**)
- ☐ Power supply through strut / slip-ring (**FA**)
- ☐ Propeller direct-drive arrangement (**FA**)
- ☐ Pod-specific FAT/testing (**FA**)
- ☐ **Electric prime-mover add-on** (Chapter 7)

16.6 Voith-Schneider (Cycloidal) Units - Complete Checklist

- ☐ VSP Unit General Assembly (**FA**)
- ☐ Blade (wing) design + blade-shaft/blade strength calculations (**FA**)
- ☐ Blade bearing arrangement (**FA**)
- ☐ Kinematic control gear / steering mechanism strength (**FA**)
- ☐ Rotor casing & main bevel gear + gear calculations (**FA**)
- ☐ Main thrust bearing design & life (**FA**)
- ☐ Bottom cover / guard plate structure (**FA**)
- ☐ Hull seat, foundation & well structure / FEA (**FA**)
- ☐ Seal system (rotor & blade shafts) (**FA**)
- ☐ Servo / hydraulic control unit (**FA**) — Chapter 10
- ☐ Steering control system & bridge console (**FA**)
- ☐ **Steering-function package** — Chapter 8 (FMEA, redundancy, SOLAS II-1/29)
- ☐ Prime-mover add-on (Chapter 7) + drive-train TVC

16.7 Water-Jet Systems - Complete Checklist

- ☐ Water-jet general arrangement & installation (**FA**)
- ☐ Impeller/rotor design & strength calculations (**FA**)
- ☐ Shaft line & bearings (**FA**) — Chapter 9
- ☐ Inlet duct & transom structure / hull penetration (**FA**)
- ☐ Steering & reversing (bucket) mechanism (**FA**)
- ☐ Steering/reversing hydraulic system (**FA**) — Chapter 10
- ☐ Control system & bridge interface (**FA**)
- ☐ Drive package (direct-from-engine or via gearbox) + TVC (**FA**)
- ☐ Performance / thrust calculations (**FA**)
- ☐ Materials, welding, NDT (**FA**)
- ☐ FAT / trials procedures (**FA**); O&M manuals (**FI**)
- ☐ If steering means of ship → Chapter 8; if DP → DP add-on

16.8 Prime-Mover Add-On Checklist (apply to any type)

If ELECTRIC-motor driven:

- ☐ Drive-motor data sheet (**FA**)
- ☐ Frequency converter (VFD) specification & arrangement (**FA**)
- ☐ Harmonic / power-quality analysis (**FA**)
- ☐ Supply transformer & switchboard interface (**FA**)
- ☐ Power cable sizing, routing & segregation (**FA**)
- ☐ Motor protection, monitoring & cooling (**FA**)

If DIESEL-engine driven:

- ☐ Diesel engine approval / type data (**FA**)
- ☐ **Torsional vibration calculation of the complete train** (**FA**)
- ☐ Clutch & flexible coupling (**FA**)
- ☐ Engine foundation & chocking (**FA**)
- ☐ Fuel oil / lube oil / cooling water systems (**FA**)
- ☐ Starting system & exhaust gas system (**FA**)
- ☐ Engine control, safety & monitoring (**FA**)
- ☐ NOx Technical File / EIAPP (**FI**)

16.9 Ice Class / Winterisation Add-On Checklist (apply if notation carried)

- ☐ Ice-class notation basis & design conditions (**FA**)
- ☐ Propeller ice-load strength calculation (**FA**)
- ☐ Shaft line & ice-torque verification (**FA**)
- ☐ Gear & azimuth/nozzle ice-load substantiation (**FA**)
- ☐ Thruster lower-unit ice protection (**FA**)
- ☐ Tunnel/inlet grid ice arrangement (**FA**)
- ☐ Winterisation of hydraulic/cooling/seal systems (**FA**)
- ☐ Low-temperature material certification (**FA**)

16.10 DP Thrusters - Additional Checklist Items

For ANY thruster type used in DP system, add:

- ☐ DP System General Arrangement (**FA**)
- ☐ Thruster Configuration Drawing (**FA**)
- ☐ DP Capability Plot (**FA**)
- ☐ FMEA Documentation (**FA**)
- ☐ Redundancy Concept (**FA**)
- ☐ Power Supply Segregation (**FA**)
- ☐ DP Control System Architecture (**FA**)
- ☐ Position Reference System Configuration (**FA**)
- ☐ PMS Configuration (**FA**)
- ☐ UPS Configuration (**FA**)
- ☐ DP FMEA Proving Trials Plan (**FA**)
- ☐ DP Trials Reports (**FI**)
- ☐ DP Operations Manual (**FI**)

Chapter 17

Common Submission Errors & Surveyor Notes

17.1 Frequent Documentation Gaps

17.1.1 Missing or Incomplete Calculations

- Shaft strength calculations missing fatigue analysis
- Bearing life calculations without proper load factors
- Gear calculations not per recognized standards (ISO 6336)
- Vibration analysis incomplete or missing
- Hydraulic system calculations (accumulator sizing, pressure drop)

17.1.2 Inadequate Drawings

- General arrangements without sufficient detail
- Material specifications not clearly indicated
- Welding details missing or incomplete
- Foundation drawings without load specifications
- Control system diagrams without protection details

17.1.3 Steering Gear Overlap Not Addressed

- Azimuth or podded thrusters used for steering without steering gear compliance documentation
- Missing FMEA for steering function
- Inadequate redundancy analysis
- Missing emergency steering provisions
- SOLAS compliance not demonstrated

17.1.4 DP System Gaps

- FMEA not covering all failure modes or worst-case scenarios
- Inadequate power segregation for DP-2/DP-3
- Missing fire/flood zone analysis for DP-3
- Thruster interaction effects not analyzed
- Incomplete proving trials procedures

17.1.5 Hydraulic System Issues

- Undersized accumulators
- Inadequate pressure relief provisions
- Missing filtration specifications
- Incomplete piping stress analysis
- Emergency provisions not clearly defined

17.1.6 Prime-Mover Gaps

- Diesel-driven Z-drive submitted *without* a torsional vibration calculation for the complete engine–clutch–shaft–thruster train (most common tug/OSV gap)
- Missing barred-speed range or clutch-in/misfiring conditions in the TVC
- Electric-driven thruster without harmonic / power-quality study or cable-sizing & segregation
- Hybrid PTI/PTO arrangements with undocumented driving paths or mode-changeover logic

17.1.7 Ice-Class & Notation Gaps

- Ice-classed vessel with propeller/shaft/gear checked only for open-water rating, not ice peak-torque / blade-milling loads
- Azimuth/pod lower unit and slewing gear not substantiated against ice-block impact
- Winterisation (low-temperature oils, materials, sensor heating) not addressed
- Comfort (COMF) notation carried but no thruster noise/vibration prediction
- Requested notation string not reconciled with the submitted thruster package

17.1.8 Cycloidal (VSP) & Combined Propulsion/Steering Gaps

- VSP mechanical package submitted without the steering-function compliance set (FMEA, redundancy, SOLAS II-1/29, response time)
- Blade guard/bottom plate and its hull attachment not substantiated for grounding/impact
- Steerable water-jet / nozzle / grid units missing steering-gear documentation

17.2 Surveyor Review Tips

17.2.1 Initial Review

1. Verify thruster power >110 kW → Section 15 applies
2. Identify thruster type → Apply appropriate checklist (Chapters 2–6)
3. Identify the prime mover (electric / diesel / hydraulic) → Add Chapter 7 submissions
4. Check for steering function → Add Chapter 8 requirements
5. Check for main shafting classification → Add Chapter 9 requirements
6. Check for DP application → Add Chapter 11 requirements
7. Check ice class / winterisation notation → Add Chapter 12 requirements
8. Reconcile the full class-notation string → Add Chapter 13 requirements

17.2.2 Calculation Review

- Verify all calculations use BV-approved formulas or recognized standards
- Check safety factors applied
- Verify material properties used in calculations match specifications
- Ensure all loading conditions considered (normal, extreme, emergency)

17.2.3 Drawing Review

- Check for consistency between drawings
- Verify dimensions and tolerances
- Ensure material specifications clearly indicated
- Check for proper welding symbols and details
- Verify accessibility for maintenance

17.3 Red Flags for Surveyors

17.3.1 Immediate Rejection Criteria

- Missing critical calculations (shaft strength, bearing life, gear strength)
- Material specifications not compliant with BV rules
- Welding procedures not approved
- Control system without adequate safety provisions
- Steering gear requirements not addressed when applicable
- DP FMEA incomplete or inadequate

17.3.2 Require Clarification

- Ambiguous material specifications
- Incomplete or unclear drawings
- Calculations without clear assumptions
- Testing procedures without acceptance criteria
- Manuals not in working language of crew

Chapter 18

Quick Reference Tables

18.1 Thruster Type vs. Applicable Rule Sections

Table 18.1: Applicability Matrix

Thruster Type	Sec.15	Sec.14	Sec.7	NR216	Hydraulics
Tunnel (Fixed)	✓	Optional	Optional	Optional	If hydraulic
Retractable	✓	Optional	Optional	Optional	✓ Required
Azimuth FP	✓	Often ✓	Typically ✓	Optional	If hydraulic
Azimuth CP	✓	Often ✓	Typically ✓	Optional	✓ Required
Podded	✓	Often ✓	Typically ✓	Optional	If CP
Water-Jet	✓	Often ✓	✓ Required	Optional	✓ Required
Voith-Schneider	✓	✓ Required	Typically ✓	Optional	✓ Required
Rim-Driven	✓	Optional	Optional	Optional	Rarely
Pump-Jet / Ducted	✓	If steering	✓ Required	Optional	If steering
Gill / Hydro-Jet	✓	Optional	Optional	Optional	If hydraulic
Steering Grid / Nozzle	✓	✓ Required	✓ Required	Optional	✓ Required

Sec.15 = Thrusters; Sec.14 = Steering Gear; Sec.7 = Shaft lines/propellers (NR467 Pt C, Ch 1).
"Often/Typically ✓" means applicable when the unit provides steering or forms part of the main shaft line.
Overlay Chapter 7 (prime mover), Chapter 12 (ice class) and Chapter 13 (notations) on every row as applicable.

18.2 Prime Mover & Additional-Notation Quick Guide

Table 18.2: Prime Mover — Key Additional Submissions

Prime mover	Headline additional submissions (see Chapter)
Electric motor	Motor data sheet; frequency converter (VFD); harmonic/power-quality study; transformer & switchboard interface; cable sizing & segregation; motor protection/monitoring/cooling.
Diesel engine	Engine approval (Sec.2); **full-train torsional vibration calculation**; clutch & flexible coupling; foundation/chocking; FO/LO/cooling/starting/exhaust systems; engine safety & controls; NOx file.
Hydraulic motor	Per Chapter 10 (HPU, circuit, accumulators, relief, filtration, monitoring).

Table 18.3: Additional Notations — Trigger Reference

Notation carried	Adds
DP / DYNAPOS	Chapter 11: FMEA, redundancy, power segregation, capability plot, proving trials (NR216).
Ice class / Polar / COLD	Chapter 12: ice-load propeller/shaft/gear strengthening, winterisation, low-temp materials (NR467 Pt E, NR527, IACS UR I3).
COMF-NOISE / COMF-VIB	Chapter 13: noise & vibration prediction, blade-rate excitation, resilient mounting.
AUT-UMS / AUT-CCS	Chapter 13: extended alarms/monitoring, safe remote control, fail-safe.
CLEANSHIP	Chapter 13: EAL seal/hydraulic fluids, leakage monitoring.
TUG / ESCORT TUG	Chapter 13: uprated steering & structural loads.

18.3 Document Approval Categories Summary

Table 18.4: Approval Categories

Category	Meaning	Timing	Surveyor Action
FA	Must be reviewed and approved before construction	Submit before construction	Detailed technical review
FI	Submitted for record, not requiring formal approval	Submit as specified	Review for completeness
During Construction	Certificates/reports generated during manufacturing	Submit as work progresses	Verify during surveys

Chapter 19

Regulatory References

19.1 Bureau Veritas Rules

19.1.1 NR467 - Rules for Classification of Steel Ships

Part B: Hull and Stability

- Chapter 2: Subdivision and Stability
- Chapter 3: Materials
- Chapter 4: Welding
- Chapter 5: Hull Structure

Part C: Machinery, Electricity, Automation and Fire Protection

- Chapter 1 — Machinery:
  - Section 2: Diesel Engines (prime-mover-driven thrusters)
  - Sections 7–8: Shaft Lines and Propellers
  - Section 14: Steering Gear
  - Section 15: Thrusters (and water-jets)
- Chapter 2 — Electrical Installations (drive motors, converters, cables)
- Chapter 3 — Automation (control, alarms, AUT notations)
- Chapter 4 — Fire Protection

Note: exact section numbering varies with rule edition — always confirm against the current NR467 in Rules Explorer.

Part D: Service Notations — e.g. tug, escort tug, anchor-handling, special purpose ship (service-driven thruster/steering loads).

Part E: Additional Class Notations

- Ice class (IA Super / IA / IB / IC) and Polar Class (PC1–PC7); COLD winterisation
- Comfort on board — COMF-NOISE, COMF-VIB
- Automated machinery spaces / integrated systems — AUT-UMS, AUT-CCS, AUT-PORT

- Dynamic positioning — DYNAPOS / DP notations
- Environmental protection — CLEANSHIP and related (EAL lubricants)
- Redundant propulsion — RP / AVM notations

19.1.2 NR216 - Rules for Dynamic Positioning Systems

- DP-1, DP-2, DP-3 Requirements
- FMEA Guidelines
- Proving Trials Requirements

19.1.3 Other BV Rule Notes (as applicable)

- **NR527** — Ships operating in polar and ice-covered waters / Polar Class
- Rule notes / guidance for comfort (noise & vibration) supporting COMF notations
- Guidance notes for cold-climate / winterisation service

19.2 International Standards

19.2.1 ISO Standards

- **ISO 6336**: Calculation of load capacity of spur and helical gears
- **ISO 22547**: Ships and marine technology — azimuth thrusters
- **ISO 484**: Shipbuilding — ship screw propellers (manufacturing tolerances)
- **ISO 4413**: Hydraulic fluid power — general rules for systems
- **ISO 20283 / ISO 6954**: Mechanical vibration — measurement & evaluation (supports COMF-VIB)

19.2.2 IEC Standards

- **IEC 60092 Series**: Electrical installations in ships
  - IEC 60092-101: Definitions and general requirements
  - IEC 60092-201: System design — general
  - IEC 60092-301: Equipment - Generators and motors
  - IEC 60092-352: Choice and installation of cables
  - IEC 60092-504: Control and instrumentation
- **IEC 60034 Series**: Rotating electrical machines (ratings, cooling, temperature rise)
- **IEC 61508**: Functional safety (for DP / thruster control systems)

19.2.3 IACS Unified Requirements & Codes

- **IACS UR I1–I3**: Polar Class — ships and machinery requirements (ice loads on propeller/shaft/gear)
- **IACS UR M**: Machinery requirements (shafting, gears, TVC principles)
- **IACS UR E22 / E26 / E27**: Programmable / cyber-resilient control systems
- **IMO Polar Code**: Ships operating in polar waters
- **Finnish-Swedish Ice Class Rules (TRAFICOM)**: Baltic ice classes basis

19.2.4 SOLAS Convention

- **Chapter II-1, Regulation 29**: Steering gear
- **Chapter II-1, Reg. 8-1 & II-2, Reg. 21/22**: Safe Return to Port / systems availability after a casualty (passenger ships)
- **Chapter II-2**: Fire protection; hazardous-area boundaries
- **Chapter V, Regulation 28**: Maneuverability

19.2.5 Ship-Type / Service Rule Sets & Codes

- **IEC 60079 series**: Explosion protection — equipment in hazardous areas (tankers, gas carriers, bow-thruster rooms)
- **IGC Code**: Gas carriers — hazardous-zone requirements
- **IMO MODU Code / offshore-unit rules**: Mobile offshore drilling units and offshore service craft
- **HSC Code**: High-speed craft (may modify/replace NR467 machinery requirements)
- **Naval ship rules** (e.g. BV NR483): Naval and patrol vessels — signature/silent operation
- **NR467 Part D**: Service notations (tug, escort tug, supply, dredging, passenger, tanker, etc.)

19.2.6 IMO Resolutions

- **IMO Res.A.468(XII)**: Code on Noise Levels
- **IMO Res.MSC.337(91)**: Code on Noise Levels on Board Ships (mandatory)
- **IMO MSC/Circ.645 & MSC.1/Circ.1580**: Guidelines for vessels/units with DP systems
- **IMO MSC.1/Circ.1369**: Interim explanatory notes for SRtP assessment

19.3 Online Resources

19.3.1 Bureau Veritas Marine & Offshore

- **Rules Explorer**: `https://rulesexplorer.bureauveritas.com/`
- **NR467 Rules**: `https://marine-offshore.bureauveritas.com/nr467-rules-classification-steel-ships`
- **All Rules & Guidance Notes**: `https://marine-offshore.bureauveritas.com/rules-classification-rule-notes-and-guidance-notes`

Chapter 20

Contact Information & Document Control

20.1 For Technical Queries

Bureau Veritas Marine & Offshore Division

- Regional Technical Support Offices
- Email: Contact through official BV channels
- Rules Explorer Support: Available through BV website

20.2 For Urgent Classification Matters

- Contact your assigned Plan Approval Manager
- Regional Marine & Offshore Director

20.3 Document Revision History

Table 20.1: Revision History

Version	Date	Changes	Author
1.0	2025	Initial comprehensive catalog based on NR467 Sections 7, 14, 15 and NR216	Jules (BV AI Assistant)
1.1	2026	Added Voith-Schneider (cycloidal) and other propulsor types (rim-driven, pump-jet, gill/jet, steering grid); prime-mover chapter (diesel-engine vs. electric-motor drive incl. full-train TVC); ice class & winterisation chapter; additional BV notations chapter (COMF, AUT, CLEANSHIP, RP, service notations); expanded checklists, quick-reference tables and standards. Preamble/compile fixes.	Plan Approval Reference (expanded ed.)
1.2	2026	Added Ship-Type & Service-Notation chapter (ship-type → thruster expectation matrix; Safe Return to Port for passenger ships; hazardous-area/Ex for tankers & gas carriers; offshore/MODU & DP-intensive vessels; tugs & escort tugs; thruster-compartment requirements). Added ship-type/service rule sets & codes to references (IEC 60079, IGC, MODU, HSC, naval, SRtP).	Plan Approval Reference (expanded ed.)

20.4 Important Disclaimers

! CRITICAL NOTICE FOR SURVEYORS:

1. **Verification Required**: This catalog is a reference guide compiled from Bureau Veritas rules. Always verify against the latest official BV rules and guidance notes.
2. **Rule Updates**: BV rules are updated periodically. Ensure you are working with the current version applicable to the vessel's build date and classification notation.
3. **Complex Cases**: For unusual thruster configurations, hybrid systems, or novel technologies, consult with BV technical specialists before approval.
4. **Project-Specific Requirements**: Some projects may have additional requirements beyond standard rules (e.g., owner specifications, flag state requirements, special notations).
5. **Overlapping Requirements**: When thrusters serve multiple functions (propulsion + steering, or DP + steering), ALL applicable requirements must be met.
6. **FMEA Critical**: For DP systems, the FMEA is the cornerstone document. Inadequate FMEA is grounds for rejection.
7. **Steering Gear Compliance**: If thrusters provide steering, full SOLAS steering gear requirements apply.
8. **Material Traceability**: All critical components must have traceable material certificates.
9. **Software Validation**: Control system software, especially for DP and steering functions, requires validation documentation.
10. **Cybersecurity**: Modern thruster control systems must address cybersecurity per latest BV guidance and IMO requirements.

! PLEASE CAREFULLY REVIEW THIS DOCUMENT

It may contain errors or omissions. Always verify against official BV rules.

Bureau Veritas

Thruster Requirements
Comprehensive Surveyor Catalog

Document Version 1.2
2026

For Bureau Veritas Plan Approval Surveyors

Source Links:
https://marine-offshore.bureauveritas.com/nr467-rules-classification-steel-ships
https://marine-offshore.bureauveritas.com/rules-classification-rule-notes-and-guidance-notes
https://rulesexplorer.bureauveritas.com/